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A robust and contact resolving Riemann solver on unstructured mesh, Part I, Euler method

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ABSTRACT

This article presents a new cell-centered numerical method for compressible flows on arbitrary unstructured meshes. A multi-dimensional Riemann solver based on the HLLC method (denoted by HLLC-2D solver) is established. The work is an extension from the cell-centered Lagrangian scheme of Maire et al. [27] to the Eulerian framework. Similarly to the work in [27], a two-dimensional contact velocity defined on a grid node is introduced, and the motivation is to keep an edge flux consistency with the node velocity connected to the edge intrinsically. The main new feature of the algorithm is to relax the condition that the contact pressures must be same in the traditional HLLC solver. The discontinuous fluxes are constructed across each wave sampling direction rather than only along the contact wave direction. The two-dimensional contact velocity of the grid node is determined via enforcing conservation of mass, momentum and total energy, and thus the new method satisfies these conservation properties at nodes rather than on grid edges. Other good properties of the HLLC-2d solver, such as the positivity and the contact preserving, are described, and the two-dimensional high-order extension is constructed employing MUSCL type reconstruction procedure. Numerical results based on both quadrilateral and triangular grids are presented to demonstrate the robustness and the accuracy of this new solver, which shows it has better performance than the existing HLLC method.

1. Introduction

Over the past decades there have been numerous investigations of the use of Riemann solvers solving multi-dimensional hyperbolic conservation laws. These methods can roughly be classified into two categories: one-dimensional Riemann solver approach and "genuinely multi-dimensional" schemes.

A natural way to solve multi-dimensional problems is the so-called "direction splitting" upwind method, which solves the governing equations in the local coordinate system attached to the interfaces between the computational cells in the normal direction. The intercell flux is determined by a one-dimensional Riemann solver. Beside exact Riemann solver for Euler flow [17] there are lots of approximate Riemann solvers, such as the linearized Riemann solver by Roe [39], the HLLE Riemann solver [20], the HLLC Riemann solvers [43,5] and some others [44].

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Although these one-dimensional Riemann solvers make success in most numerical calculations, some researchers have always believed that the one-dimensional Riemann solvers lose much of their efficacy in multi-dimensional problems. Indeed in multi-dimensional flow, acoustic waves can propagate in infinitely many directions rather than just two, and vorticity exists as a multi-dimensional phenomenon. The one-dimensional upwind method cannot completely account for all flow features that might be propagating transverse to the cell boundary. Some Riemann solvers, which can resolve the contact and shear waves exactly such as Roe [39] and HLLC [43] solvers, may suffer from "carbuncle" and "odd–even decoupling" phenomena that are called the numerical shock instability [38].

One remedy to this defect is to combine different types of flux functions into so-called hybrid Riemann solvers: one is sharp in capturing discontinuities and the other is dissipative but stable for multi-dimensional shocks [38]. Such approaches are very useful but involve a user-defined parameter which is set to determine when and where to use the hybrid Riemann solver [35,21,22,36].

The necessity to incorporate genuinely multi-dimensional physics into upwind algorithms was recognized and studied by Roe [40]. From then, numerous solutions have been proposed to introduce multi-dimensional features into CFD solvers. Wendroff [46], Kurganov and Tadmor [24], Kurganov and Petrova [25] and, more recently, Balsara [1–3], Capdeville [8] developed multi-dimensional extension of the HLL or HLLC scheme. Brio et al. [6] generalized the one-dimensional Roe method to two-dimensional case, Guinot [19] proposed a two-dimensional solver based on angular Riemann problem. In these methods multistate Riemann initial values are input, and the configurations of the wave interaction are considered. Some complex wave interactions are simplified or averaged. Among them some methods [25,8,6,19] and [3] are suited for triangular or arbitrary unstructured grids, and other ones are only restricted to Cartesian grids due to their elaborate construction [46,24,33]. By now, the advantages of genuinely multi-dimensional schemes are quite remarkable, such as Balsara et al.'s methods [1–3] permit larger CFL to be used and the flow features show greater isotropy in multi-dimensional flows than those associated with one-dimensional Riemann solver. Not only that, their methods have been extended to other conservation laws such as MHD flows.

In multi-dimensional Riemann solvers, Després and Mazeran [12] and Maire et al. [27] propose two-dimensional Riemann nodal solvers in Lagrangian framework respectively. Such solvers introduce explicitly a vertex contact velocity and keep the consistency between the nodal contact velocity and normal moving velocity of an edge connected to the node. The advantages of the methods are to enhance the stability of the Lagrangian computation compared with the traditional method used in CAVEAT code [13,14]. Their methods are suit for arbitrary unstructured meshes and can trace material interface exactly due to the Lagrangian property. The numerical results in the above papers show that they are quite robust and accurate. From then, a series of contributions to the development of the method are performed. Burton et al. device a new multi-direction Riemann nodal solver for solid dynamics [7], and Morgan et al. apply their nodal solver to construct a contact surface algorithm [31]. Carré et al. extend Després and Mazeran's method to arbitrary dimension [9]. Maire develops high-order schemes on two-dimensional Cartesian and cylindrical geometries [28,29]. More work can be found in [11,26] and [30].

By now Maire et al.'s method has been extended to ALE method by Lagrange plus remap algorithm. However this solver provides physical quantities only on particle motion locus and the method has not yet been extended to direct Eulerian method. This is in part because their nodal algorithm has discontinue pressure terms cross an edge and such discontinue fluxes bring a lot difficulties when discretizing advection terms in equations, which is different from those finite volume methods based on one-dimensional Riemann solver.

In this paper, we present a new cell-centered Eulerian scheme to solve the compressible gas dynamics equations on arbitrary two-dimensional unstructured meshes. The main work is to construct a two-dimensional Riemann solver which has one-dimensional flux form and includes multi-dimensional information. Similar to the work of Maire et al.'s method, a two-dimensional particle flow velocity defined at grid vertex is introduced, and the edge flux need to keep the consistency with the nodal contact velocity, although the grid does not move with the fluids. The nodal contact velocity is determined by the conservation principles of mass, momentum and total energy, therefore the new method satisfies these conservation properties at nodes rather than on grid edges. For the flux formulation, we use the HLLC solver instead of acoustic wave solver in order to suit for the Eulerian calculation. The solver relaxes the condition that the contact pressures must be same in the traditional HLLC solver, and in contrast with Maire et al.'s method, its main new feature lies in providing fluxes in arbitrary wave propagation direction rather than only in contact wave propagation direction. Because the resulting numerical flux makes use of the information in all neighboring cells, the solver is a kind of genuinely two-dimensional Riemann solver and will be called as HLLC-2D.

The two-dimensional high-order extension is constructed by employing MUSCL-like reconstruction procedure. Numerical tests are provided to demonstrate the good performance such as the robustness and the accuracy of this new solver by comparisons with competing numerical methods.

Many researchers have reported failures of the approximate Riemann solvers capable of capturing contact discontinuity accurately in the presence of strong shock [18,10]. It is worth to point out that the new method can eliminate such numerical instability very well, moreover it is a contact preserving method.

The outline of this paper is as follows. In Section 2 the governing equations are given. A general framework of numerical scheme is presented in Section 3 and the traditional HLLC solver is introduced. The new Eulerian method and its properties are described in Sections 4 and 5 respectively. A high order extension to new scheme is given in Section 6. The corresponding numerical results in Section 7 provide a clear evidence of the robustness of this new scheme, especially compared with the traditional HLLC scheme. Finally, the conclusions are summarized in Section 8.

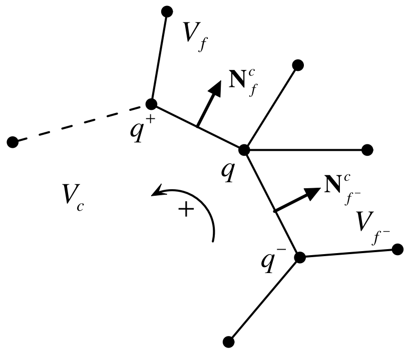

Fig. 1. Notations on a generic polygonal grid.

2. Governing equations

The governing equations of inviscid flow in two-dimension are as follows:

$$\frac{\partial \mathbf{U}}{\partial t} + \frac{\partial \mathbf{F}(\mathbf{U})}{\partial x} + \frac{\partial \mathbf{G}(\mathbf{U})}{\partial y} = 0, \tag{1}$$

where the state vector and flux vectors are

$$\mathbf{U} = \begin{bmatrix} \rho \\ \rho u \\ \rho v \\ \rho E \end{bmatrix}, \qquad \mathbf{F}(\mathbf{U}) = \begin{bmatrix} \rho u \\ \rho u^2 + p \\ \rho u v \\ \rho E u + p u \end{bmatrix}, \qquad \mathbf{G}(\mathbf{U}) = \begin{bmatrix} \rho v \\ \rho u v \\ \rho v^2 + p \\ \rho E v + p v \end{bmatrix},$$

where ρ, p, E are the fluid density, pressure and total energy respectively, $\mathbf{u} = (u, v)$ is the fluid velocity. The equation of state is in the form of

$$p = (\gamma - 1)\rho e = (\gamma - 1)\rho\left(E - \frac{1}{2}(u^2 + v^2)\right),$$

where γ is the specific heat ratio.

It is convenient, for the subsequent discretization, to recast the system (1) of equations in the following moving control volume formulation [13]:

$$\frac{d}{dt}\int_{\Omega} \mathbf{U}\,dx\,dy + \int_{\partial\Omega} \big[(\mathbf{F}, \mathbf{G}) \cdot \mathbf{N} - (\mathbf{w} \cdot \mathbf{N})\mathbf{U}\big]\,dl = 0, \tag{2}$$

where $\mathbf{w}$ is the moving velocity of the control volume Ω, and $\mathbf{N}$ is the unit outward normal direction on the boundary of Ω. If $\mathbf{w} = \mathbf{u}$, the system reduces to a Lagrangian form and if $\mathbf{w} = 0$, the equations have Eulerian form.

3. Discrete scheme in space

3.1. Notations on a generic polygonal grid

Each cell V_c of the mesh is assigned a unique index c. We use the index f to denote a generic neighboring cell V_f which has common edge with cell V_c, and the edge is denoted as $c \wedge f$, referring to Fig. 1. $\mathcal{F}(c)$ is the set of such neighboring cells around cell V_c. Each vertex of the mesh is assigned a unique index q, and $\mathcal{C}(q)$ and $\mathcal{K}(q)$ are defined to be the sets of cells and edges respectively around the vertex q, i.e.,

$$\mathcal{C}(q) = \{V_c\colon \text{ cells sharing a vertex } q\},$$

$$\mathcal{K}(q) = \{k\colon \text{ edges sharing a vertex } q\}.$$

The physical quantities such as the density ρ_c, pressure p_c, velocity $\mathbf{u}_c$, energy E_c, e_c are defined in the cell center of V_c. In the notation of unit normal vector $\mathbf{N}_f^c$, the subscript f and the superscript c describe the end and start points of the vector.

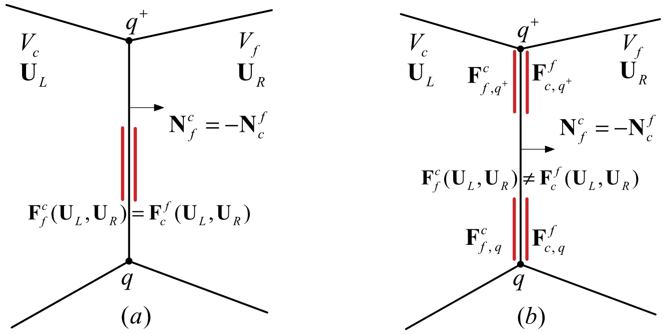

Fig. 2. Comparison between a traditional Eulerian method based on one-dimensional Riemann solver (a) and the new method with HLLC-2D solver (b). Fluxes across an edge of the cells can be different in the new method whereas they are the same in the traditional method.

We discrete these equations by utilizing the idea of the Godunov scheme on unstructured mesh. A finite-volume scheme for Eq. (2) is obtained by considering the control volume balance equation

$$\begin{aligned} \mathbf{U}_c^{n+1} &= \mathbf{U}_c^n - \frac{\Delta t}{|V_c|} \sum_{f \in \mathcal{F}(c)} \int_f (\mathcal{T}_f^c)^{-1} \mathbf{F}(\mathcal{T}_f^c \mathbf{U})\, dl \\ &\approx \mathbf{U}_c^n - \frac{\Delta t}{|V_c|} \sum_{f \in \mathcal{F}(c)} L_f^c (\mathcal{T}_f^c)^{-1} \mathbf{F}_f^c(\mathcal{T}_f^c \mathbf{U}_c, \mathcal{T}_f^c \mathbf{U}_f), \end{aligned} \tag{3}$$

where $|V_c|$ denotes the volume of cell V_c, and $\Delta t = t^{n+1} - t^n$. $\mathbf{N}_f^c = ((n_x)_f^c, (n_y)_f^c)$ is the unit outward norm of the edge $c \wedge f$ between V_c and V_f and L_f^c is the length of the edge. $\mathcal{T}_f^c$ is the rotation matrix and $(\mathcal{T}_f^c)^{-1}$ is its inverse, namely

$$\mathcal{T}_f^c = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & (n_x)_f^c & (n_y)_f^c & 0 \\ 0 & -(n_y)_f^c & (n_x)_f^c & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}, \qquad (\mathcal{T}_f^c)^{-1} = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & (n_x)_f^c & -(n_y)_f^c & 0 \\ 0 & (n_y)_f^c & (n_x)_f^c & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}. \tag{4}$$

The numerical fluxes $\mathbf{F}_f^c$ between the cells V_c and V_f can be obtained from many methods. The most popular way is to solve a one-dimensional Riemann problem along the normal direction of the edge $c \wedge f$,

$$\begin{aligned} &\mathbf{U}_t + \mathbf{F}_x = 0, \\ &IVS\colon \quad \mathbf{U}(x, 0) = \begin{cases} \mathbf{U}_L = \mathcal{T}_f^c \mathbf{U}_c^n, & x < 0, \\ \mathbf{U}_R = \mathcal{T}_f^c \mathbf{U}_f^n, & x > 0. \end{cases} \end{aligned} \tag{5}$$

In the above notations, the superscript c is used to emphasize the fact these quantities are viewed from cell V_c. From cell V_f, see Fig. 2, there are

$$L_c^f = L_f^c, \qquad \mathbf{N}_c^f = -\mathbf{N}_f^c. \tag{6}$$

In a cell centered finite volume method, most of numerical fluxes are defined at middle point of edge $c \wedge f$, and satisfy the following consistent condition,

$$(\mathcal{T}_c^f)^{-1} \mathbf{F}_f^c(\mathcal{T}_c^f \mathbf{U}_f, \mathcal{T}_c^f \mathbf{U}_c) = -(\mathcal{T}_f^c)^{-1} \mathbf{F}_f^c(\mathcal{T}_f^c \mathbf{U}_c, \mathcal{T}_f^c \mathbf{U}_f), \tag{7}$$

and the conservation condition across a cell face

$$\mathbf{F}_c^f(\mathbf{U}_L, \mathbf{U}_R) = \mathbf{F}_f^c(\mathbf{U}_L, \mathbf{U}_R). \tag{8}$$

In such cases, the conservation of mass, momentum and total energy are kept naturally, and it's unnecessary to use index c or f to distinguish between the two fluxes.

It is well known that a wide variety of approximate Riemann solvers have been proposed to solve the above Riemann problem. These solvers are much cheaper than exact Riemann solver and give results that in many cases are equally good when used in Godunov methods. A familiar example is the HLLC approximate solver.

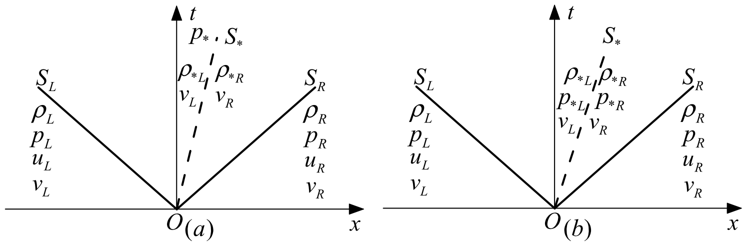

Fig. 3. Comparison between the HLLC and HLLC-2D approximate Riemann solver. Pressures in the Star Region are discontinuous in the HLLC-2D (b) whereas they are the same in the HLLC solver (a).

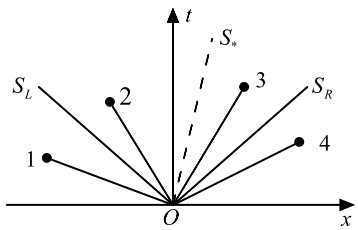

Fig. 4. Four possible cases of location of the segment in the wave pattern.

3.2. The HLLC approximate solver

The HLLC approximate solver includes 4 states and can accurately capture contact and shear waves for x-split two-dimensional gas dynamics, see Fig. 3. It is initially proposed by Toro, Spruce, and Speares [43] to overcome the flaw of HLL approximate Riemann [20] which cannot distinguish isolated contact discontinuities. The state vector is

$$\mathbf{U}_{HLLC}(\mathbf{U}_L, \mathbf{U}_R) = \begin{cases} \mathbf{U}_L, & \text{if } 0 \leqslant S_L, \\ \mathbf{U}_L^*, & \text{if } S_L < 0 \leqslant S_*, \\ \mathbf{U}_R^*, & \text{if } S_* < 0 \leqslant S_R, \\ \mathbf{U}_R, & \text{if } 0 > S_R, \end{cases} \tag{9}$$

where $\mathbf{U}_K^*$ for $K = L, R$ satisfy

$$\mathbf{U}_K^* = \rho_K \frac{S_K - u_K}{S_K - S_*} \begin{bmatrix} 1 \\ S_* \\ v_K \\ E_K + (S_* - u_K)(S_* + \frac{p_K}{\rho_K(S_K - u_K)}) \end{bmatrix}, \tag{10}$$

and S_L, S_R are the wave speeds on either side. Once given the wave speed estimations, the contact speed S_* is available in terms of S_L, S_R,

$$S_* = \frac{p_L - p_R - (\rho u)_L(S_L - u_L) + (\rho u)_R(S_R - u_R)}{(\rho u)_L - (\rho u)_R + S_R\rho_R - S_L\rho_L}. \tag{11}$$

Notice that there is a useful relation between $\mathbf{U}$ and $\mathbf{F}$, namely

$$\mathbf{F}(\mathbf{U}) = u\mathbf{U} + \mathbf{D}, \quad \mathbf{D} = p(0, 1, 0, u)^T.$$

The numerical interface flux (referring to Fig. 4) is

$$\mathbf{F}_{HLLC}(\mathbf{U}_L, \mathbf{U}_R) = \begin{cases} \mathbf{U}_L u_L + \mathbf{D}_L, & \text{if } 0 \leqslant S_L, \\ \mathbf{U}_L^* S_* + \mathbf{D}^*, & \text{if } S_L < 0 \leqslant S_*, \\ \mathbf{U}_R^* S_* + \mathbf{D}^*, & \text{if } S_* < 0 \leqslant S_R, \\ \mathbf{U}_R u_R + \mathbf{D}_R, & \text{if } 0 > S_R, \end{cases} \tag{12}$$

where $\mathbf{D}_K = p_K(0, 1, 0, u_K)^T$, and $\mathbf{D}^* = p^*(0, 1, 0, S_*)^T$ for $K = L, R$.

Given S_L, S_R, the pressure p^* in the Star Region ($S_L < x/t < S_R$) is

$$p^* = p_L + \rho_L(S_L - u_L)(S_* - u_L) = p_R + \rho_R(S_R - u_R)(S_* - u_R). \tag{13}$$

In order to complete the HLLC Riemann solver, S_R and S_L must be estimated appropriately. There are many choices to obtain them [44]. Here we use those proposed by Einfeld et al. [15] to estimate waves speeds

$$S_L = \min(u_L - c_L, \bar{u} - \bar{c}), \qquad S_R = \max(u_R + c_R, \bar{u} + \bar{c}), \tag{14}$$

where c_L and c_R are the sound speeds of the left and the right states, $\bar{u}$, $\bar{c}$ are the relevant Roe average quantities [44].

We can also adopt the adaptive approximate-state Riemann solver presented by Toro [44], or a simpler estimate in [5],

$$S_L = \min(u_L - c_L, u_R - c_R), \qquad S_R = \max(u_L + c_L, u_R + c_R). \tag{15}$$

Obviously, these wavespeed estimates satisfy

$$S_L \leqslant u_L - c_L < u_L, \qquad S_R \geqslant u_R + c_R > u_R. \tag{16}$$

The HLLC method can resolve the contact wave exactly for any choice of the wave velocities, and can resolve accurately isolated shock when using a special wavespeed estimate (14) [5]. It satisfies the consistency condition (7) and the conservation condition (8), moreover, the approach has the positively conservative property [5]. However, when calculating two-dimensional strong shock problems, some numerical shock instability phenomena may appear [21]. It is because the perturbation transfer in the transverse direction of shock waves cannot be well restrained. A hybrid method including more dissipative HLL flux and HLLC flux functions can cure this defect [38,21], but the property of the contact preserving is lost. A two-dimensional extension of the HLLC Riemann solver based on a nodal solution can reduce the numerical shock instability which is the main topic of this paper.

4. A new Eulerian method

The new Eulerian method has a close relation with a set of cell centered Lagrangian ones with nodal solvers. To clearly demonstrate the motivation to design the new algorithm, we present a review of the cell centered Lagrangian methods.

4.1. A review of the Lagrangian approaches

Different from the Eulerian method, the Lagrangian method tracks the fluid motion and is free of advection errors. This means the mesh moves with fluids as time goes on.

The finite volume discrete scheme for the Lagrangian equations (2) can be written as follows

$$\begin{aligned}
\rho_c^{n+1} &= \frac{|V_c^n|}{|V_c^{n+1}|}\rho_c^n, \\
\rho_c^{n+1}\mathbf{u}_c^{n+1} &= \frac{|V_c^n|}{|V_c^{n+1}|}\rho_c^n\mathbf{u}_c^n - \frac{\Delta t}{|V_c^{n+1}|}\sum_f L_f^c p_f^c \mathbf{N}_f^c, \\
\rho_c^{n+1}E_c^{n+1} &= \frac{|V_c^n|}{|V_c^{n+1}|}\rho_c^n E_c^n - \frac{\Delta t}{|V_c^{n+1}|}\sum_f L_f^c p_f^c \nu_f^c,
\end{aligned} \tag{17}$$

which is a discrete analogue to (3) in the circumstance of $\mathbf{w} = \mathbf{u}$. In Eq. (17), p_f^c, ν_f^c are the contact pressure and the face-normal component of the contact velocity. $|V_c^n|$ and $|V_c^{n+1}|$ are volumes of cell V_c at time t^n and t^{n+1}. The node position at t^{n+1} can be obtained by

$$\mathbf{x}_q^{n+1} = \mathbf{x}_q^n + \Delta t \mathbf{u}_q^*, \tag{18}$$

where $\mathbf{u}_q^*$ is the contact velocity allocated at the vertex q.

There exists two different discrete forms for the cell volume. One is consistent with the flux form in (17),

$$\left|V_c^{n+1}\right| = \left|V_c^n\right| + \Delta t \sum_f L_f^c \nu_f^c. \tag{19}$$

Another is from the node displacement and has the following approximate form

$$\left|V_c^{n+1}\right| = \left|V_c^n\right| + \frac{1}{2}\Delta t \sum_f L_f^c \left(\mathbf{u}_q^* + \mathbf{u}_{q^+}^*\right) \cdot \mathbf{N}_f^c. \tag{20}$$

In the early cell-centered Godunov-type method used in the CAVEAT code [13], the cell face quantities p_f^c, ν_f^c are available from the solution of a one-dimensional approximate Riemann problem in the direction normal to the face $c \wedge f$.

When determining a vertex contact velocity $\mathbf{u}_q^*$ to move the grid, a question arises. One cannot find such a vertex velocity $\mathbf{u}_q^*$ that its normal projection on any of the connecting edges is equal to the corresponding contact velocity ν_f^c

on the edge simultaneously. In fact, this solution procedure forms an over-determined system. The CAVEAT algorithm in [13] uses a weighted least squares algorithm to overcome this difficulty and obtain the velocity $\mathbf{u}_q^*$. The algorithm leads to inconsistency between the scheme (19) and scheme (20), which means incompatibility between a discrete geometric conservation and the nodal motion.

Després and Mazeran [12], and Maire et al. [27] notice that the flux computation was not compatible with the node displacement in the Lagrangian CAVEAT algorithm [13]. They regard that such inconsistency may generate additional spurious components in the vertex velocity field. Indeed, there exists severe vorticity errors in the CAVEAT algorithm [14].

An important achievement concerning the compatibility between flux discretization and vertex velocity computation has been introduced in [12,27]. In these papers, the authors present an innovational idea in which the contact pressures across the cell face are not necessarily equal and the local conservation is satisfied by the combined effects of all cells around a vertex.

For example, in [27], four pressures are introduced on each edge, two for each node on each side of the edge, see Fig. 2(b). The fluxes in (17) cross the face $c \wedge f$ are

$$p_f^c = \frac{1}{2}\left(p_{f,q}^c + p_{f,q^+}^c\right), \tag{21}$$

$$p_f^c \nu_f^c = \frac{1}{2}\left(p_{f,q}^c \mathbf{u}_q^* \cdot \mathbf{N}_f^c + p_{f,q^+}^c \mathbf{u}_{q^+}^* \cdot \mathbf{N}_f^c\right), \tag{22}$$

where the pressures $p_{f,q}^c$ and p_{f,q^+}^c satisfy the acoustic wave approximate relations

$$\begin{aligned} p_{f,q}^c &= p_c + \rho_c c_c\left(\mathbf{u}_q^* \cdot \mathbf{N}_f^c - \mathbf{u}_c \cdot \mathbf{N}_f^c\right), \\ p_{f,q^+}^c &= p_c + \rho_c c_c\left(\mathbf{u}_{q^+}^* \cdot \mathbf{N}_f^c - \mathbf{u}_c \cdot \mathbf{N}_f^c\right). \end{aligned} \tag{23}$$

In above formulae, c_c is the sound speed on the cell V_c, $\mathbf{u}_q^*$ is the contact velocity at the vertex q, whose normal projection on the face $c \wedge f$ is the face normal contact velocity $\nu_{f,q}^c$. $\mathbf{u}_q^*$ is available from the local conservation of the mass, momentum and total energy around the node q.

Adopting this solution procedure, the two volume discrete formulations (19) and (20) keep the equivalency. The numerical fluxes have the coherent manner with the node motion. The numerical results shown in [12] and [27] are quite impressive. From then, a series of contributions to the development of the method are performed. Burton et al. device a new multi-direction Riemann nodal solver for solid dynamics [7], and Morgan et al. apply the nodal solver in [7] to construct a contact surface algorithm [31]. Carré et al. extend Després and Mazeran's method to arbitrary dimension [9], and Maire develops a high-order scheme on two-dimensional Cartesian and cylindrical geometries [28,29]. More work can be found in [30,16].

However, little progress is made in extending the two-dimensional cell-centered Lagrangian schemes to the Eulerian and the arbitrary Lagrangian Eulerian framework. One still uses the Lagrangian step plus remapping method [26]. This is because the implementation of the advection terms including a discontinuous pressure and work term is rather difficult.

In our new Eulerian method, although the grid no longer moves, the nodal contact velocity $\mathbf{u}_q^*$ still need to be introduced explicitly to guarantee the intrinsic compatibility between the flow at the vertex and the edge-normal contact velocity.

4.2. A new Eulerian method

The new method is to construct a two-dimensional Riemann solver HLLC-2D. It defines explicitly a nodal contact velocity $\mathbf{u}_q^*$ at each grid vertex q and introduces two half-edge fluxes $\mathbf{F}_{f,q}^c, \mathbf{F}_{f,q^+}^c$ on each edge $c \wedge f$ from the cell V_c, see Fig. 2(b). The final face flux $\mathbf{F}_f^c$ satisfies

$$\mathbf{F}_f^c\left(\mathcal{T}_f^c\mathbf{U}_c, \mathcal{T}_f^c\mathbf{U}_f\right) = \frac{1}{2}\left(\mathbf{F}_{f,q}^c\left(\mathcal{T}_f^c\mathbf{U}_c, \mathcal{T}_f^c\mathbf{U}_f\right) + \mathbf{F}_{f,q^+}^c\left(\mathcal{T}_f^c\mathbf{U}_c, \mathcal{T}_f^c\mathbf{U}_f\right)\right). \tag{24}$$

If the edge flux is viewed from the cell V_f, then other two edge-face fluxes $\mathbf{F}_{c,q}^f$, $\mathbf{F}_{c,q^+}^f$ on the edge $c \wedge f$ are introduced. Therefore there are four half-edge fluxes totally on the edge.

In order to clearly describe the construction of a new Eulerian method, let us introduce some new notations shown in Fig. 5. We denote by q a generic internal vertex of the mesh and $\mathcal{C}(q)$ the set of cells that surround point q. Cells V_{c^-}, V_c and V_{c^+} are in the counterclockwise list of $\mathcal{C}(q)$. The left state L and the right state R (between a red line) of a cell face k are determined according to the normal direction of the face, and they are used to solve a Riemann problem.

The algorithm includes two key ingredients. One is a nodal solver which is based on a vertex q with multi-state initial values $\{\mathbf{U}_c,\ c \in \mathcal{C}(q)\}$ around the vertex q. The nodal contact velocity $\mathbf{u}_q^*$ will be evaluated from this solver. Another is an extension of the classical one-dimensional HLLC Riemann solver, and we call it HLLC-2D solver. It will provide an approximate flux formula for $F_{c^+}^c$ after introducing the nodal velocity. That means such approximate flux depends on the initial states $\mathbf{U}_c$, $\mathbf{U}_{c^+}$ and $\mathbf{u}_q^*$.

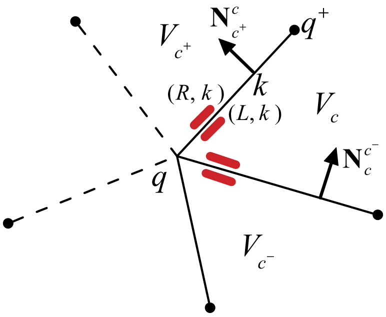

Fig. 5. States around vertex q. Riemann solver HLLC-2D is built on the states between each edge. The edge $c \wedge c^+$ between the cells V_c and V_{c^+} is sometimes denoted as k, and the left state and the right state is determined according to the direction of the normal vector of the edge k. (For interpretation of the colors in this figure, the reader is referred to the web version of this article.)

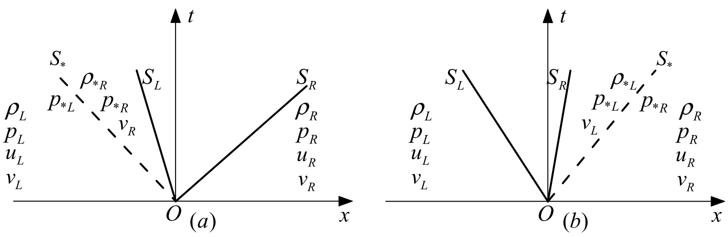

Fig. 6. The possible wave configurations of the solver HLLC-2D due to the projection of the nodal contact velocity.

4.3. The flux formula of the HLLC-2D solver

In this subsection, we focus on the flux formula cross an edge of the Riemann solver HLLC-2D on the assumption that the contact velocity $\mathbf{u}_q^*$ at a vertex has been determined. To keep the consistency of description, let's return to Fig. 2(b).

Although the Riemann problem belongs to a typical two-dimensional case, the solution still can be expressed as a special approximate form for the one-dimensional Riemann problem (5).

4.3.1. The structure of the approximate solver

According to the normal direction of the cell edge, the initial values of the one-dimensional Riemann problem (5) between the edge $c \wedge f$ can be denoted by the left and the right data:

$$\mathbf{U}_L = \mathcal{T}_f^c \mathbf{U}_c, \qquad \mathbf{U}_R = \mathcal{T}_f^c \mathbf{U}_f,$$

and the corresponding fluxes are $\mathbf{F}_L = \mathbf{F}(\mathbf{U}_L)$, $\mathbf{F}_R = \mathbf{F}(\mathbf{U}_R)$.

Similar to the HLLC solver, the new approximate solution to the Riemann problem consists of a contact wave S_* and two acoustic waves S_L and S_R. In most of instances, two averaged states $\mathbf{U}_L^*, \mathbf{U}_R^*$ are separated by the contact wave. The difference from the HLLC solver is that the contact discontinuity $S_* = \mathbf{u}_q^* \cdot \mathbf{N}_f^c$ is not calculated by the one-dimensional jump relations. Due to the two-dimensional characters, sometimes the contact velocity S_* is on the left or the right of the signal velocity boundaries (S_L, S_R), see Fig. 6. Notice that the contact velocity S_* is not derived from the one-dimensional Rankine–Hugoniot relations, the contact pressure on this edge seen from L may be different of that seen from R. We denote the contact pressures as p_L^* and p_R^*.

Now, we are going to show how to compute the states in star regions and interface fluxes on the $x/t = 0$. A Lagrangian description is used to obtain start states, and two one-side Rankine–Hugoniot relations are employed to obtain the fluxes.

4.3.2. A Lagrangian description

It is well known that any flow of a fluid always can be interpreted by either the Lagrangian or the Eulerian viewpoint. The Lagrangian description may be the simplest way to calculate $\mathbf{U}_L^*$, $\mathbf{U}_R^*$ and p_L^*, p_R^*.

In the Lagrangian frame, S_* is a velocity boundary and the states in the start regions are quantities on the boundary. They satisfy the following Rankine–Hugoniot conditions

$$\begin{bmatrix} \rho_K^* S_* \\ \rho_K^* S_*^2 + p_K^* \\ \rho_K^* S_* v_K \\ S_*(\rho_K^* E_K^* + p_K^*) \end{bmatrix} - \begin{bmatrix} \rho_K u_K \\ \rho_K u_K^2 + p_K \\ \rho_K u_K v_K \\ u_K(\rho_K E_K + p_K) \end{bmatrix} = S_K \begin{bmatrix} \rho_K^* \\ \rho_K^* S_* \\ \rho_K^* v_K \\ \rho_K^* E_K^* \end{bmatrix} - S_K \begin{bmatrix} \rho_K \\ \rho_K u_K \\ \rho_K v_K \\ \rho_K E_K \end{bmatrix} \tag{25}$$

where $K = L, R$.

From (25) we may write the following solutions for the pressures on the contact wave

$$p_K^* = p_K + \rho_K(S_K - u_K)(S_* - u_K), \tag{26}$$

and the state vectors $\mathbf{U}_L^*$, $\mathbf{U}_R^*$

$$\mathbf{U}_K^* = \frac{S_K \mathbf{U}_K - \mathbf{F}_K + \mathbf{D}_K^*}{S_K - S_*}, \tag{27}$$

where $K = L, R$ and $\mathbf{D}_K^* = p_K^*(0, 1, 0, S_*)^T$.

Notice that the $\mathbf{U}_L^*, \mathbf{U}_R^*$ have the same formulae as (10) although the different contact pressures exist.

Remark 1. If we implement the contact pressure (26) to a half edge $c \wedge f$ connected to the node q in Fig. 2(b) (denote p_L^* as $p_{f,q}^c$) and remain the Lagrangian approach, the resulting Lagrangian scheme (17) with (21) and (22) can be regarded as an extension of the Maire et al.'s method [27], in which the acoustic wave solver in (23) is replaced by the HLLC solver (26). In other words, the Maire et al.'s method is a special case of the HLLC solver, where wave speed estimate adopts the acoustic approximations,

$$S_L = u_L - c_L, \qquad S_R = u_R + c_R, \tag{28}$$

where c_L, c_R are the sound speeds on left and right states. It is well known that the wave velocity estimation (28) may result in false $S_L > S_R$ in the presence of strong shock.

We stress that introducing two pressures to a half edge is reasonable, it does not break the physical sanctity of the contact discontinuity. The local balance can be attained even if there exists the pressure difference on a half edge.

Let us return to Fig. 5. Under the combined interaction of all cells around the node q, the local force balance is

$$\sum_{k \in \mathcal{K}(q)} L_k \left(p_{L,k}^* - p_{R,k}^*\right)\mathbf{N}_k = 0, \tag{29}$$

where L_k is the length of the edge k, and $p_{L,k}^*$ and $p_{R,k}^*$ correspond to the pressures in (26). $\mathcal{K}(q)$ is the set of edges sharing the vertex q.

Substituting (26) into (29) and making use of $S_* = \mathbf{u}_q^* \cdot \mathbf{N}_k$, the nodal contact velocity $\mathbf{u}_q^*$ can be calculated from (29). The Lagrangian solution procedure is completely same as that in [27].

Since the contact velocity S_* is the projection of the node velocity $\mathbf{u}_q^*$ to the edge $c \wedge f$, it has possibility to locate on the outside of the range (S_L, S_R), see Fig. 6. This is a new character of two-dimensional flows although it is difficult to be interpreted in a one-dimensional x–t plot. On the assumption that the node has unique velocity, these phenomena may appear in rotation flow or shear flow. In these cases, $p_K^*, \mathbf{U}_K^*$ still can be calculated via formulae (26) and (27), and (26) keeps a qualitative correctness. Let's discuss it in detail.

Under the circumstance in Fig. 6(a), the velocity boundary moves to the left and the left state is compressed, at the same time the right state becomes expansion. The pressure formula (26) validates the intuitionistic conclusion. In fact, according to (16), there is

$$S_* < S_L < u_L, \qquad S_* < S_L < u_R < S_R.$$

It is easy to deduce from (26) that $p_L^* > p_L$ and $p_R^* < p_R$. Notice that at this time $\mathbf{U}_R^*$ represents a star state but $\mathbf{U}_L^*$ loses physical meaning because

$$\rho_R^* = \rho_R \frac{S_R - u_R}{S_R - S_*} > 0, \qquad \rho_L^* = \rho_L \frac{S_L - u_L}{S_L - S_*} < 0.$$

Similar case appears in Fig. 6(b), the right state is compressed and the left state is expanded. At this time, $\rho_L^* > 0$, $\rho_R^* < 0$, therefore only the left star state $\mathbf{U}_L^*$ has meaning.

In the Lagrangian calculation, one needs not the information of $\mathbf{U}_K^*$ for $K = L, R$. The numerical results obtained from our method [42] and Maire et al.'s approach [27] show that such nonphysical middle star states do not cause any problems even if there exists strong shock.

4.3.3. *The approximate Riemann fluxes*

It is time to consider the advection fluxes in Eulerian framework. Our approximate Riemann solver is based on above defined states. Two approximate fluxes are introduced and correspond to left and right states. They ensure the jump conditions are satisfied strictly when the fluid passes through discontinuous line between different states.

At the beginning we assume the inequality $S_L \leqslant S_* \leqslant S_R$ holds, see Fig. 3. Define four fluxes from left to right as $\mathbf{F}_{L,k}$, $\mathbf{F}_{R,k}$, $k = 1, 2, 3, 4$, (refer to Fig. 4). The Rankine–Hugoniot relations across the S_L, S_*, S_R waves give the following equations,

$$
\begin{aligned}
&\mathbf{F}_{L,2}-\mathbf{F}_{L,1}=S_L\big(\mathbf{U}_L^*-\mathbf{U}_L\big)=\mathbf{F}_{R,2}-\mathbf{F}_{R,1},\\
&\mathbf{F}_{L,3}-\mathbf{F}_{L,2}=S_*\big(\mathbf{U}_R^*-\mathbf{U}_L^*\big)=\mathbf{F}_{R,3}-\mathbf{F}_{R,2},\\
&\mathbf{F}_{L,4}-\mathbf{F}_{L,3}=S_R\big(\mathbf{U}_R-\mathbf{U}_R^*\big)=\mathbf{F}_{R,4}-\mathbf{F}_{R,3},
\end{aligned}
\tag{30}
$$

where the boundary fluxes are $\mathbf{F}_{L,1}=\mathbf{F}_L$, $\mathbf{F}_{R,4}=\mathbf{F}_R$ respectively.

After performing certain algebraic manipulations of these equations, one obtains the following solutions

$$
\begin{cases}
\mathbf{F}_{L,1}=\mathbf{F}_L, & \mathbf{F}_{R,1}=\mathbf{F}_L-\mathbf{D}_L^*+\mathbf{D}_R^*,\\
\mathbf{F}_{L,2}=\mathbf{U}_L^*S_*+\mathbf{D}_L^*, & \mathbf{F}_{R,2}=\mathbf{U}_L^*S_*+\mathbf{D}_R^*,\\
\mathbf{F}_{L,3}=\mathbf{U}_R^*S_*+\mathbf{D}_L^*, & \mathbf{F}_{R,3}=\mathbf{U}_R^*S_*+\mathbf{D}_R^*,\\
\mathbf{F}_{L,4}=\mathbf{F}_R-\mathbf{D}_R^*+\mathbf{D}_L^*. & \mathbf{F}_{R,4}=\mathbf{F}_R.
\end{cases}
\tag{31}
$$

Then the numerical fluxes at the interface are one side ones

$$
\hat{\mathbf{F}}_L(\mathbf{U}_L,\mathbf{U}_R)=
\begin{cases}
\mathbf{U}_L u_L+\mathbf{D}_L, & \text{if } 0\leqslant S_L,\\
\mathbf{U}_L^*S_*+\mathbf{D}_L^*, & \text{if } S_L<0\leqslant S_*,\\
\mathbf{U}_R^*S_*+\mathbf{D}_L^*, & \text{if } S_*<0\leqslant S_R,\\
\mathbf{U}_R u_R+\mathbf{D}_R-\mathbf{D}_R^*+\mathbf{D}_L^*, & \text{if } 0>S_R.
\end{cases}
\tag{32}
$$

$$
\hat{\mathbf{F}}_R(\mathbf{U}_L,\mathbf{U}_R)=
\begin{cases}
\mathbf{U}_L u_L+\mathbf{D}_L-\mathbf{D}_L^*+\mathbf{D}_R^*, & \text{if } 0\leqslant S_L,\\
\mathbf{U}_L^*S_*+\mathbf{D}_R^*, & \text{if } S_L<0\leqslant S_*,\\
\mathbf{U}_R^*S_*+\mathbf{D}_R^*, & \text{if } S_*<0\leqslant S_R,\\
\mathbf{U}_R u_R+\mathbf{D}_R, & \text{if } 0>S_R.
\end{cases}
\tag{33}
$$

Such approximate fluxes have a clear physical interpretation. For brevity, we discuss one case: $S_L<S_*<0\leqslant S_R$, and other cases can be explained similarly.

For a flow which has the contact pressure p_L^* and p_R^*, it has a motion to the left since $S_*<0\leqslant S_R$. At a very short time, the convective part passing cross the fixed edge (which has the sampling direction $x/t=0$) is $\mathbf{U}_R^*$, and flow velocity is S_*. Thus during a unit time, the advection term that the left state obtain and the right states lose is $\mathbf{U}_R^*S_*$. The whole two fluxes correspond to the third case in (32) and (33).

Next, let's consider two special wave configurations in Fig. 6. If the contact velocity $S_*<S_L<S_R$ (Fig. 6(a)), then similar to the above discussion, we have

$$
\hat{\mathbf{F}}_L(\mathbf{U}_L,\mathbf{U}_R)=\mathbf{U}_R^*S_*+\mathbf{D}_L^*,\qquad \hat{\mathbf{F}}_R(\mathbf{U}_L,\mathbf{U}_R)=\mathbf{U}_R^*S_*+\mathbf{D}_R^*.
$$

If $S_L<S_R<S_*$ (referring to Fig. 6(b)), then

$$
\hat{\mathbf{F}}_L(\mathbf{U}_L,\mathbf{U}_R)=\mathbf{U}_L^*S_*+\mathbf{D}_L^*,\qquad \hat{\mathbf{F}}_R(\mathbf{U}_L,\mathbf{U}_R)=\mathbf{U}_L^*S_*+\mathbf{D}_R^*.
$$

Remark 2. For the two new wave configurations in the HLLC-2D solver, the nonphysical middle states do not be used. In practical calculation, the two special cases are classified into (32) and (33), and need not be specially treated.

Notice that the numerical fluxes depend on the contact wave velocity S_* of the vertex q, the interface flux seen from the cell V_c in the HLLC-2D solver can be expressed by

$$
\mathbf{F}_{f,q}^c(\mathbf{U}_L,\mathbf{U}_R)=\hat{\mathbf{F}}_L(\mathbf{U}_L,\mathbf{U}_R).
\tag{34}
$$

The interface flux to the cell V_f is

$$
\mathbf{F}_{c,q}^f(\mathbf{U}_L,\mathbf{U}_R)=\hat{\mathbf{F}}_R(\mathbf{U}_L,\mathbf{U}_R),
\tag{35}
$$

where we have added the subscript q to represent the flux related with node q.

When viewing the edge $c\wedge f$ from V_f, two numerical fluxes $\bar{\mathbf{F}}_{c,q}^f(\bar{\mathbf{U}}_L,\bar{\mathbf{U}}_R)$ and $\bar{\mathbf{F}}_{f,q}^c(\bar{\mathbf{U}}_L,\bar{\mathbf{U}}_R)$ can be defined similarly as (34) and (35). In the two fluxes, the normal direction and local coordinate system change,

$$
\mathbf{N}_c^f=-\mathbf{N}_f^c,\qquad \bar{\mathbf{U}}_L=\mathcal{T}_c^f\mathbf{U}_f,\qquad \bar{\mathbf{U}}_R=\mathcal{T}_c^f\mathbf{U}_c.
$$

The following claim shows that the fluxes $\bar{\mathbf{F}}_{c,q}^f$ and $\bar{\mathbf{F}}_{f,q}^c$ are consistent with $\mathbf{F}_{c,q}^f$ and $\mathbf{F}_{f,q}^c$ respectively.

Claim 1. *The fluxes (34) and (35) satisfy consistency relations (7),*

$$
\big(\mathcal{T}_c^f\big)^{-1}\bar{\mathbf{F}}_{c,q}^f\big(\mathcal{T}_c^f\mathbf{U}_f,\mathcal{T}_c^f\mathbf{U}_c\big)=-\big(\mathcal{T}_f^c\big)^{-1}\mathbf{F}_{c,q}^f\big(\mathcal{T}_f^c\mathbf{U}_c,\mathcal{T}_f^c\mathbf{U}_f\big),
\tag{36}
$$

$$
\big(\mathcal{T}_c^f\big)^{-1}\bar{\mathbf{F}}_{f,q}^c\big(\mathcal{T}_c^f\mathbf{U}_f,\mathcal{T}_c^f\mathbf{U}_c\big)=-\big(\mathcal{T}_f^c\big)^{-1}\mathbf{F}_{f,q}^c\big(\mathcal{T}_f^c\mathbf{U}_c,\mathcal{T}_f^c\mathbf{U}_f\big).
\tag{37}
$$

Proof. Notice that

$$\bar{\mathbf{U}}_L = \begin{bmatrix} \rho_R \\ -\rho_R u_R \\ -\rho_R v_R \\ \rho_R E_R \end{bmatrix}, \qquad \bar{\mathbf{U}}_R = \begin{bmatrix} \rho_L \\ -\rho_L u_L \\ -\rho_L v_L \\ \rho_L E_L \end{bmatrix},$$

there is the flux according to Eq. (33),

$$\bar{\mathbf{F}}^f_{c,q}(\bar{\mathbf{U}}_L, \bar{\mathbf{U}}_R) = \begin{cases} \bar{\mathbf{U}}_L \bar{u}_L + \bar{\mathbf{D}}_L, & \text{if } 0 \leqslant \bar{S}_L, \\ \bar{\mathbf{U}}^*_L \bar{S}_* + \bar{\mathbf{D}}^*_L, & \text{if } \bar{S}_L < 0 \leqslant \bar{S}_*, \\ \bar{\mathbf{U}}^*_R \bar{S}_* + \bar{\mathbf{D}}^*_L, & \text{if } \bar{S}_* < 0 \leqslant \bar{S}_R, \\ \bar{\mathbf{U}}_R \bar{u}_R + \bar{\mathbf{D}}_R - \bar{\mathbf{D}}^*_R + \bar{\mathbf{D}}^*_L, & \text{if } 0 > \bar{S}_R. \end{cases}$$

The velocities and signal speeds in this local coordinate system are opposite to those with initial values $(\mathbf{U}_L, \mathbf{U}_R)$,

$$\bar{u}_L = -u_R, \qquad \bar{S}_L = -S_R, \qquad \bar{u}_R = -u_L, \qquad \bar{S}_R = -S_L, \qquad \bar{S}_* = -S_*.$$

Other states have the same signals,

$$\bar{\rho}^*_L = \rho^*_R, \qquad \bar{p}^*_L = p^*_R, \qquad \bar{E}^*_L = E^*_R, \qquad \bar{\rho}^*_R = \rho^*_L, \qquad \bar{p}^*_R = p^*_L, \qquad \bar{E}^*_R = E^*_L.$$

Without loss of generalization, we only consider the case $S_L < 0 \leqslant S_*$, i.e., $\bar{S}_* \leqslant 0 < \bar{S}_R$. At this time, the flux is

$$\bar{\mathbf{F}}^f_{c,q}(\bar{\mathbf{U}}_L, \bar{\mathbf{U}}_R) = \bar{\mathbf{U}}^*_R \bar{S}_* + \bar{\mathbf{G}}^*_L = \begin{bmatrix} \rho^*_L \\ -\rho^*_L S_* \\ -\rho^*_L v_L \\ \rho^*_L E^*_L \end{bmatrix} (-S_*) + \begin{bmatrix} 0 \\ p^*_R \\ 0 \\ -p^*_R S_* \end{bmatrix} = \begin{bmatrix} -\rho^*_L S_* \\ \rho^*_L S_* S_* + p^*_R \\ \rho^*_L v_L S_* \\ -(\rho^*_L E^*_L S_* + p^*_R S_*) \end{bmatrix}.$$

Note that in case $S_L < 0 \leqslant S_*$ the flux (34) has the following form:

$$\mathbf{F}^f_{c,q}(\mathbf{U}_L, \mathbf{U}_R) = \mathbf{U}^*_L S_* + \mathbf{D}^*_R = \begin{bmatrix} \rho^*_L \\ \rho^*_L S_* \\ \rho^*_L v_L \\ \rho^*_L E^*_L \end{bmatrix} S_* + \begin{bmatrix} 0 \\ p^*_R \\ 0 \\ p^*_R S_* \end{bmatrix} = \begin{bmatrix} \rho^*_L S_* \\ \rho^*_L S_* S_* + p^*_R \\ \rho^*_L v_L S_* \\ \rho^*_L E^*_L S_* + p^*_R S_* \end{bmatrix},$$

thus there holds

$$\left(\mathcal{T}^f_c\right)^{-1} \bar{\mathbf{F}}^f_{c,q}(\bar{\mathbf{U}}_L, \bar{\mathbf{U}}_R) = -\left(\mathcal{T}^c_f\right)^{-1} \mathbf{F}^f_{c,q}(\mathbf{U}_L, \mathbf{U}_R).$$

Likewise,

$$\left(\mathcal{T}^f_c\right)^{-1} \bar{\mathbf{F}}^c_{f,q}(\bar{\mathbf{U}}_L, \bar{\mathbf{U}}_R) = -\left(\mathcal{T}^c_f\right)^{-1} \mathbf{F}^c_{f,q}(\mathbf{U}_L, \mathbf{U}_R).$$

We finish the proof of Claim 1. □

Due to the consistency between $\bar{\mathbf{F}}^f_{c,q}$ and $\mathbf{F}^f_{c,q}$, we can drop off the overbar '–' of $\bar{\mathbf{F}}^f_{c,q}$, which will not lead to any confusion.

In addition, there are two obvious properties for the solver HLLC-2D.

Claim 2.

$$\mathbf{F}^c_{f,q}(\mathbf{U}_L, \mathbf{U}_R) - \mathbf{F}^f_{c,q}(\mathbf{U}_L, \mathbf{U}_R) = \mathbf{D}^*_L - \mathbf{D}^*_R. \tag{38}$$

This property shows the flux difference cross an edge still keeps the contact pressure difference. It will be used in the calculation of the nodal contact velocity and display the consistency with the Lagrangian approach.

Claim 3. *If S_* satisfies (11), then the Riemann solver HLLC-2D reduces to the traditional HLLC solver.*

Similar derivation can be done at vertex q^+, and fluxes $\mathbf{F}^c_{f,q^+}$, $\mathbf{F}^f_{c,q^+}$ are obtained. The resulting numerical fluxes on $c \wedge f$ are

$$\mathbf{F}^c_f(\mathbf{U}_L, \mathbf{U}_R) = \frac{1}{2}\big[\mathbf{F}^c_{f,q}(\mathbf{U}_L, \mathbf{U}_R) + \mathbf{F}^c_{f,q^+}(\mathbf{U}_L, \mathbf{U}_R)\big], \tag{39}$$

$$\mathbf{F}^f_c(\mathbf{U}_L, \mathbf{U}_R) = \frac{1}{2}\big[\mathbf{F}^f_{c,q}(\mathbf{U}_L, \mathbf{U}_R) + \mathbf{F}^f_{c,q^+}(\mathbf{U}_L, \mathbf{U}_R)\big]. \tag{40}$$

In summary, the new method includes two characteristics different from traditional one:

1. Introduce a unique contact velocity $\mathbf{u}^*_q$ at each grid vertex q.
2. Define different fluxes $\mathbf{F}^c_{f,q}$, $\mathbf{F}^c_{f,q^+}$, $\mathbf{F}^f_{c,q}$, $\mathbf{F}^f_{c,q^+}$ on each edge, two for each node on each side of the edge. The edge flux $\mathbf{F}^c_f$ is no longer equal to $\mathbf{F}^f_c$ necessarily due to introduced contact discontinuity.

4.4. Conservation relations

Since there exists discontinuity between two fluxes $\mathbf{F}^c_f$ and $\mathbf{F}^f_c$ on the interface of cells V_c and V_f, the counteraction of two fluxes no longer holds and the conservation of scheme is not obvious. To keep the conservation of mass, momentum and energy, it needs to make a global balance of discrete equations, and replace the global summation over cells by a global summation over nodes.

Let's return to Fig. 5. Omitting the boundary conditions and summing equations on all cells of the domain, there is

$$\begin{aligned}
\frac{d}{dt}\int_\Omega \mathbf{U}\,dx\,dy &\approx \sum_c \frac{\mathbf{U}^{n+1}_c - \mathbf{U}^n_c}{\Delta t}|V_c| \\
&= -\sum_c \sum_{f\in\mathcal{F}(c)} L^c_f\big(\mathcal{T}^c_f\big)^{-1}\mathbf{F}^c_f\big(\mathcal{T}^c_f\mathbf{U}_c, \mathcal{T}^c_f\mathbf{U}_f\big) \\
&= -\frac{1}{2}\sum_q \sum_{c\in\mathcal{C}(q)} \big[L^c_{c^+}\big(\mathcal{T}^c_{c^+}\big)^{-1}\mathbf{F}^c_{c^+,q}\big(\mathcal{T}^c_{c^+}\mathbf{U}_c, \mathcal{T}^c_{c^+}\mathbf{U}_{c^+}\big) + L^c_{c^-}\big(\mathcal{T}^c_{c^-}\big)^{-1}\mathbf{F}^c_{c^-,q}\big(\mathcal{T}^c_{c^-}\mathbf{U}_c, \mathcal{T}^c_{c^-}\mathbf{U}_{c^-}\big)\big] \\
&\equiv: -\frac{1}{2}\Big(\sum_{q\in I(\Omega)}\sum_{c\in\mathcal{C}(q)} \mathbf{FF}_{c,q} + \sum_{q\in\mathcal{B}(\Omega)}\sum_{c\in\mathcal{C}(q)} \mathbf{FF}_{c,q}\Big),
\end{aligned} \tag{41}$$

where $\mathcal{B}(\Omega)$ is the set of boundary nodes, and $\mathcal{I}(\Omega)$ represents the set of interior nodes. Here we have replaced the global summation over cells by a summation over nodes. $\mathbf{FF}_{c,q}$ are the outward normal fluxes cross the edges of cell V_c connected to the vertex q, see Fig. 5. The discrete conservation law requires

$$\sum_{q\in\mathcal{I}(\Omega)}\sum_{c\in\mathcal{C}(q)} \mathbf{FF}_{c,q} = 0.$$

A sufficient condition to preserve the conservation is

$$\sum_{c\in\mathcal{C}(q)} \mathbf{FF}_{c,q} = 0, \quad \text{if } q \in I(\Omega). \tag{42}$$

Notice that

$$\begin{aligned}
\sum_{c\in\mathcal{C}(q)} \mathbf{FF}_{c,q} &= \sum_{c\in\mathcal{C}(q)} L^c_{c^+}\big[\big(\mathcal{T}^c_{c^+}\big)^{-1}\mathbf{F}^c_{c^+,q}\big(\mathcal{T}^c_{c^+}\mathbf{U}_c, \mathcal{T}^c_{c^+}\mathbf{U}_{c^+}\big) + \big(\mathcal{T}^{c^+}_c\big)^{-1}\mathbf{F}^{c^+}_{c,q}\big(\mathcal{T}^{c^+}_c\mathbf{U}_{c^+}, \mathcal{T}^{c^+}_c\mathbf{U}_c\big)\big] \\
&= \sum_{c\in\mathcal{C}(q)} L^c_{c^+}\big(\mathcal{T}^c_{c^+}\big)^{-1}\big[\mathbf{F}^c_{c^+,q}\big(\mathcal{T}^c_{c^+}\mathbf{U}_c, \mathcal{T}^c_{c^+}\mathbf{U}_{c^+}\big) - \mathbf{F}^{c^+}_{c,q}\big(\mathcal{T}^c_{c^+}\mathbf{U}_c, \mathcal{T}^c_{c^+}\mathbf{U}_{c^+}\big)\big]
\end{aligned} \tag{43}$$

This summation is done over all the cells V_c surrounding the vertex q. A shift from $c^- \wedge c$ to $c^+ \wedge c$ is performed and the last equality uses Claim 1. A sufficient condition for conservation is

$$\sum_{c\in\mathcal{C}(q)} L^c_{c^+}\big(\mathcal{T}^c_{c^+}\big)^{-1}\big[\mathbf{F}^c_{c^+,q}\big(\mathcal{T}^c_{c^+}\mathbf{U}_c, \mathcal{T}^c_{c^+}\mathbf{U}_{c^+}\big) - \mathbf{F}^{c^+}_{c,q}\big(\mathcal{T}^c_{c^+}\mathbf{U}_c, \mathcal{T}^c_{c^+}\mathbf{U}_{c^+}\big)\big] = 0. \tag{44}$$

We denote by k the edge between the cells V_c and V_{c^+}. Denote the counterclockwise unit normal $\mathbf{N}^c_{c^+}$ on the edge k is $\mathbf{N}_k$ and its length is L_k. In this circumstance, V_c and V_{c^+} are left and right cells of edge k respectively. Notice that from Claim 2,

$$\mathbf{F}^c_{c^+,q}(\mathbf{U}_{L,k}, \mathbf{U}_{R,k}) - \mathbf{F}^{c^+}_{c,q}(\mathbf{U}_{L,k}, \mathbf{U}_{R,k}) = \mathbf{D}^*_{L,k} - \mathbf{D}^*_{R,k}.$$

Substituting the above formula into (44), we get

$$\sum_{k\in\mathcal{K}(q)} L_k\big(p^*_{L,k}-p^*_{R,k}\big)\mathbf{N}_k=0, \tag{45}$$

and

$$\sum_{k\in\mathcal{K}(q)} L_k\big(p^*_{L,k}-p^*_{R,k}\big)\mathbf{N}_k\cdot\mathbf{u}^*_q=0, \tag{46}$$

where $\mathcal{K}(q)$ is the set of edges sharing the vertex q.

Once the condition (45) holds, which implies (46) holds, the local mass, momentum and total energy conservations will be preserved. Notice that (45) is just the same with (29), which is the conservation condition of the Lagrangian approach. This means that the Eulerian HLLC-2D method has inherent consistency with the Lagrangian HLLC-2D one.

We will evaluate the $\mathbf{u}^*_q$ in a consistent way to make the scheme keeping locally conservation of mass, momentum and total energy.

4.5. Nodal solver

After the substitution of (26) into (45), we obtain the following equation

$$\sum_{k\in\mathcal{K}(q)} L_k(\alpha_{L,k}+\alpha_{R,k})\big[\mathbf{u}^*_q\cdot\mathbf{N}_k-\nu^*_k\big]\mathbf{N}_k=0, \tag{47}$$

where

$$\alpha_{L,k}=-\rho_{L,k}(S_{L,k}-u_{L,k}),\qquad \alpha_{R,k}=\rho_{R,k}(S_{R,k}-u_{R,k}), \tag{48}$$

and ν^*_k is the normal velocity given by the classical one-dimensional HLLC Riemann solver (11) for edge k

$$\nu^*_k=\frac{p_{L,k}-p_{R,k}+\alpha_{L,k}u_{L,k}+\alpha_{R,k}u_{R,k}}{\alpha_{L,k}+\alpha_{R,k}}. \tag{49}$$

We solve Eq. (47) to obtain the contact velocity $\mathbf{u}^*_q$ at each vertex q. There are

$$\mathbf{u}^*_q=\mathcal{M}^{-1}\sum_{k\in\mathcal{K}(q)} L_k(\alpha_{L,k}+\alpha_{R,k})\nu^*_k\mathbf{N}_k, \tag{50}$$

where

$$\mathcal{M}=\begin{bmatrix}\sum_{k\in\mathcal{K}(q)} L_k(\alpha_{L,k}+\alpha_{R,k})n^2_{x,k} & \sum_{k\in\mathcal{K}(q)} L_k(\alpha_{L,k}+\alpha_{R,k})n_{x,k}n_{y,k}\\ \sum_{k\in\mathcal{K}(q)} L_k(\alpha_{L,k}+\alpha_{R,k})n_{x,k}n_{y,k} & \sum_{k\in\mathcal{K}(q)} L_k(\alpha_{L,k}+\alpha_{R,k})n^2_{y,k}\end{bmatrix}.$$

If the acoustic approximation is adopted in the wave velocity estimation,

$$S_{L,k}=u_{L,k}-c_{L,k},\qquad S_{R,k}=u_{R,k}-c_{R,k},$$

where $c_{L,k}$, $c_{R,k}$ are the sound speeds on left and right states, then the nodal solver reduces to Maire et al.'s method [27].

4.6. Boundary conditions

In the calculations using Eulerian formalism, there are many types of boundary conditions, such as periodic, nonreflecting, solid wall, inflow and outflow boundary conditions. The latter three cases can be categorized to a boundary type with normal component of the velocity.

Although our Godunov-type method needs to determine nodal contact velocity $\mathbf{u}^*_q$, but we still use zero-order extrapolation method to implement boundary conditions. For velocity boundaries, we need to distinguish two cases. (1) When three boundary nodes are not collinear, we only introduce two ghost cells to implement simple extrapolation like that in Fig. 7(a). (2) When three boundary nodes are collinear, we implement mirror extrapolation, i.e., a symmetrical ghost point is introduced for each vertex connecting a boundary node, referring to Fig. 7(b).

Let's see an example with outflow boundary, refer to Fig. 7. Suppose ν_1 and ν_{K+1} be the values of the normal outflow velocities on the boundary edges coming out of q. The two edges are boundary of cells V_{c_1} and V_{c_K}. One needs to introduce

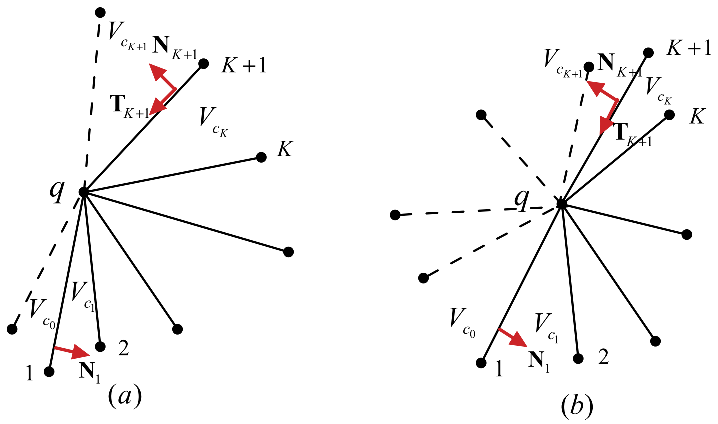

Fig. 7. Notations for the boundary conditions. When boundary nodes 1, q, $K+1$ are not collinear, two ghost cells are introduced. When the three nodes are collinear, the mirror symmetrical ghost cells are introduced.

states on a ghost cell $V_{c_{K+1}}$ to build a one-dimensional Riemann problem. If the state on cell V_{c_K} is

$$\mathbf{U}_L = \begin{bmatrix} \rho_{c_K} \\ \rho_{c_K}\mathbf{u}_{c_K}\cdot\mathbf{N}_{K+1} \\ \rho_{c_K}\mathbf{u}_{c_K}\cdot\mathbf{T}_{K+1} \\ (\rho e)_{c_K} + \frac{1}{2}\rho_{c_K}\mathbf{u}_{c_K}\cdot\mathbf{u}_{c_K} \end{bmatrix},$$

where $\mathbf{T}_{K+1}$ is unit tangent vector on edge $K+1$. Then state on the ghost cell is

$$\mathbf{U}_R = \begin{bmatrix} \rho_{c_K} \\ \rho_{c_K}(2\nu_{K+1} - \mathbf{u}_{c_K}\cdot\mathbf{N}_{K+1}) \\ \rho_{c_K}\mathbf{u}_{c_K}\cdot\mathbf{T}_{K+1} \\ (\rho e)_{c_K} + \frac{1}{2}\rho_{c_K}[(2\nu_{K+1} - \mathbf{u}_{c_K}\cdot\mathbf{N}_{K+1})^2 + (\mathbf{u}_{c_K}\cdot\mathbf{T}_{K+1})^2] \end{bmatrix}.$$

4.7. Time step limitation

Maire et al. [27] propose a time steplength condition to guaranty heuristically the monotone behavior of the entropy and ensure a positive entropy production in cell V_c. This limitation is very reliable in moving mesh method but is too strict to fixed grid method. Here we still use the standard CFL criterion.

At time t^n, for each cell V_c we denote by λ_c the radius of inscribed circle of the cell. We define

$$\Delta t = C_E \min_c \frac{\lambda_c}{\sqrt{u_c^2 + v_c^2} + c_c} \tag{51}$$

where C_E is a strictly positive coefficient and c_c is the sound speed in the cell. Extensive numerical experiments show that $C_E = 0.3$ is a value which provides good numerical results.

5. Some properties of the HLLC-2D solver

5.1. The genuinely multi-dimensional nature

There are some fundamental differences between our method associated to the HLLC-2D solver and the scheme based on a one-dimensional Riemann solver. Except more wave configurations, another difference can be illustrated by the consideration of a mesh made of rectangular cells. We denote by 1, 2, 3 and 4 the vertices of the cell V_c and 1–8 the neighboring cells, see Fig. 8. In a standard finite volume scheme with a one-dimensional Riemann solver, the cell V_c only exchanges information with the neighboring cells having a common face, i.e. the cells V_1–V_4. It yields a 5 point scheme.

With our scheme, the contact velocity at the vertex M_2 is obtained from (50),

$$\mathbf{u}_2^* = \begin{pmatrix} \frac{L_{M_5M_2}(\alpha_{L,M_5M_2}+\alpha_{R,M_5M_2})\nu^*_{M_5M_2}+L_{M_2M_3}(\alpha_{L,M_2M_3}+\alpha_{R,M_2M_3})\nu^*_{M_2M_3}}{L_{M_5M_2}(\alpha_{L,M_5M_2}+\alpha_{R,M_5M_2})+L_{M_2M_3}(\alpha_{L,M_2M_3}+\alpha_{R,M_2M_3})} \\ \frac{L_{M_2M_1}(\alpha_{L,M_2M_1}+\alpha_{R,M_2M_1})\nu^*_{M_2M_1}+L_{M_9M_2}(\alpha_{L,M_9M_2}+\alpha_{R,M_9M_2})\nu^*_{M_9M_2}}{L_{M_2M_1}(\alpha_{L,M_2M_1}+\alpha_{R,M_2M_1})+L_{M_9M_2}(\alpha_{L,M_9M_2}+\alpha_{R,M_9M_2})} \end{pmatrix} \tag{52}$$

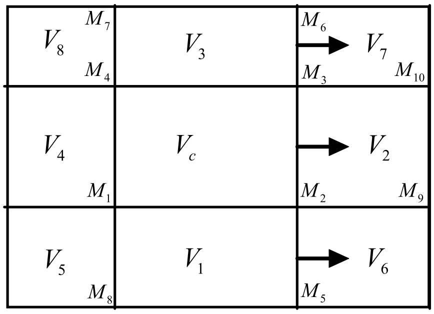

Fig. 8. Stencil and flux expression for a rectangular mesh.

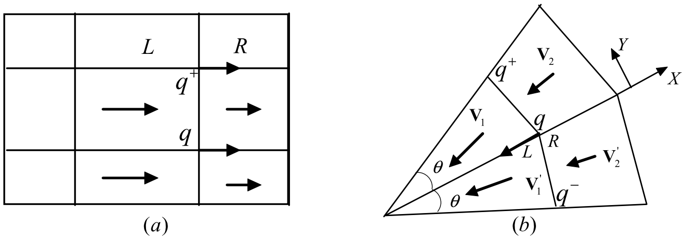

Fig. 9. One-dimensional planar and cylindrical flow.

where $v^*_{M_5M_2}$ is the one-dimensional HLLC contact velocity between cells V_1 and V_6. The other notations may refer to Fig. 8.

The projection of the contact velocity $\mathbf{u}^*_2$ to the edge M_2M_3 is

$$S_{*,M_2M_3,2} = \frac{L_k(\alpha_{L,k}+\alpha_{R,k})v^*_k|_{k=M_5M_2} + L_k(\alpha_{L,k}+\alpha_{R,k})v^*_k|_{k=M_2M_3}}{L_k(\alpha_{L,k}+\alpha_{R,k})|_{k=M_5M_2} + L_k(\alpha_{L,k}+\alpha_{R,k})|_{k=M_2M_3}}$$

Similarly, the projection of $\mathbf{u}^*_3$ to the edge M_2M_3 is

$$S_{*,M_2M_3,3} = \frac{L_k(\alpha_{L,k}+\alpha_{R,k})v^*_k|_{k=M_3M_6} + L_k(\alpha_{L,k}+\alpha_{R,k})v^*_k|_{k=M_2M_3}}{L_k(\alpha_{L,k}+\alpha_{R,k})|_{k=M_3M_6} + L_k(\alpha_{L,k}+\alpha_{R,k})|_{k=M_2M_3}}$$

Substituting these expressions of S_* into (32) to obtain the numerical fluxes on the edge M_2M_3, the information of cells V_6 and V_7 are involved, and the projected contact velocity on M_2M_3 is the average of M_5M_2, M_2M_3 and M_3M_6. The resulting update of cell V_c needs all the neighboring cells having a common vertex, and make the HLLC-2D solver be a 9 point scheme including the cells V_1–V_8. The four additional cells reinforce the genuinely multi-dimensional nature.

5.2. Two degenerate one-dimensional flows

For one-dimensional flow with planar and cylindrical symmetry, it is easy to prove that the solver will recover exactly the HLLC Riemann solver when using certain special symmetrical meshes, see Fig. 9.

In fact, in the case of a one-dimensional planar flow with rectangular mesh (Fig. 9(a)), the flow states across the edge qq^+ are denoted L and R respectively. The flow velocities are $\mathbf{u}_L = (u_L, 0)$, $\mathbf{u}_R = (u_R, 0)$ in the left and right cells of the edge qq^+, we can solve the nodal contact velocity from (52) as

$$\mathbf{u}^*_q = \left(\frac{p_L - p_R - (\rho u)_L(S_L - u_L) + (\rho u)_R(S_R - u_R)}{(\rho u)_L - (\rho u)_R + S_R\rho_R - S_L\rho_L}, 0\right)^T.$$

In the circumstances of a one-dimensional cylindrical symmetry flow with an equiangular mesh, we use the coordinate system in Fig. 9(b), and the flow velocities around a vertex q are

$$\mathbf{V}_1 = u_L\left(\cos\frac{\theta}{2}, \sin\frac{\theta}{2}\right)^T, \qquad \mathbf{V}_2 = u_R\left(\cos\frac{\theta}{2}, \sin\frac{\theta}{2}\right)^T,$$

$$\mathbf{V}'_1 = u_L\left(\cos\frac{\theta}{2}, -\sin\frac{\theta}{2}\right)^T, \qquad \mathbf{V}'_2 = u_R\left(\cos\frac{\theta}{2}, -\sin\frac{\theta}{2}\right)^T.$$

One can obtain the contact velocity from (50) [27]

$$\mathbf{u}_q^* = \left(\frac{p_L - p_R - (\rho u)_L (S_L - u_L) + (\rho u)_R (S_R - u_R)}{(\rho u)_L - (\rho u)_R + S_R \rho_R - S_L \rho_L} \frac{1}{\cos \frac{\theta}{2}}, 0 \right)^T.$$

In these instances, the projection of $\mathbf{u}_q^*$ to the edge connecting vertex q will degenerate to be contact velocity of the one-dimensional HLLC solver.

5.3. Exact resolution of isolated contact wave

If the flow is uniform (i.e. with a uniform velocity $\mathbf{u}_0$ and pressure p_0), we can easily check that the velocity at any vertex q reduces to $\mathbf{u}_q^* = \mathbf{u}_0$.

Let's return to Fig. 5 and consider the nodal velocity formula (50). In this case,

$$p_{L,k} = p_{R,k} = p_0, \qquad u_{L,k} = u_{R,k} = \mathbf{u}_0 \cdot \mathbf{N}_k.$$

Therefore from (49), we have $v_k^* = \mathbf{u}_0 \cdot \mathbf{N}_k$, and from (50),

$$\mathbf{u}_q^* = \mathcal{M}^{-1} \sum_{k \in \mathcal{K}(q)} L_k (\alpha_{L,k} + \alpha_{R,k}) \mathbf{u}_0 \cdot \mathbf{N}_k \mathbf{N}_k = \mathbf{u}_0.$$

The projection velocity on the edge k is $S_* = \mathbf{u}_q^* \cdot \mathbf{N}_k$. According to Claim 3, the HLLC-2D solver reduces to the HLLC one oncemore. It can resolute the isolated contacts exactly for any choice of the signal speeds in (15) and (14).

5.4. Positivity preservation

According to [15], a Riemann solver yields a positively conservative scheme iff all the states generated are physically real. This set of physically realistic states is defined as follows,

$$G = \left\{ \begin{bmatrix} \rho \\ \rho u \\ \rho v \\ \rho E \end{bmatrix}, \ \rho > 0, \ E - \frac{1}{2}(u^2 + v^2) > 0 \right\}.$$

When $\mathbf{U}_L \in G$, $\mathbf{U}_R \in G$, we require that all intermediate states $\mathbf{U}_L^*$, $\mathbf{U}_R^*$ satisfy the above positive condition, that is

$$\rho_K^* > 0, \qquad e_K^* > 0. \tag{53}$$

From (27) and (10), we know

$$\rho_K^* = \frac{\rho_K (S_K - u_K)}{S_K - S_*}. \tag{54}$$

(1) If $S_L < S_* < S_R$, then $\rho_K^* > 0$ always holds.
(2) If $S_* < S_L < S_R$ (see Fig. 6(a)), then $\mathbf{U}_L^*$ does not appear in the HLLC-2D solver, and $\rho_R^* > 0$.
(3) If $S_L < S_R < S_*$ (see Fig. 6(b)), then $\mathbf{U}_R^*$ does not appear in the HLLC-2D solver, and $\rho_L^* > 0$.

From (10), there is

$$\begin{aligned} e_K^* &= E_K + (S_* - u_K)\left(S_* + \frac{p_K}{\rho_K (S_K - u_K)}\right) - \frac{1}{2}(S_*^2 + v_K^2) \\ &= e_K + \frac{1}{2} u_K^2 - \frac{1}{2} S_*^2 + (S_* - u_K)\left(S_* + \frac{p_K}{\rho_K (S_K - u_K)}\right) \\ &= e_K + \frac{1}{2}(S_* - u_K)^2 + \frac{p_K (S_* - u_K)}{\rho_K (S_K - u_K)}. \end{aligned} \tag{55}$$

The sufficient condition to keep $e_K^* > 0$ is to let the discriminant of the above quadratic with respective to $(S_* - u_K)$ be negative.

$$\begin{aligned} \Delta &= \frac{p_K^2}{(S_K - u_K)^2} - 2\rho_K^2 e_K = \frac{p_K^2}{(S_K - u_K)^2} - \frac{2 p_K \rho_K}{\gamma - 1} \\ &= \frac{p_K \rho_K c_K^2}{\gamma (S_K - u_K)^2} - \frac{2 p_K \rho_K}{\gamma - 1} = \frac{p_K \rho_K}{\gamma}\left(\frac{c_K^2}{(S_K - u_K)^2} - \frac{2\gamma}{\gamma - 1}\right). \end{aligned}$$

Obviously, if

$$|S_K - u_K| > \sqrt{\frac{\gamma-1}{2\gamma}} c_K, \tag{56}$$

then $\Delta < 0$, and $e_K^* > 0$.

Notice the fact $\gamma > 1$. It follows that both the wavespeed estimates in (15) and (14) can guarantee the inequality (56) holds. Therefore the solver HLLC-2D preserves the positivity property of the density and the specific inner energy.

6. MUSCL-type finite volume methods on unstructured grids

To construct a high order numerical scheme, we employ some classical reconstruction techniques and the limiting function. For each component of the state variable vector, the limited reconstruction within a cell is expressed as follows,

$$W_c(\mathbf{x}) = \bar{W}_c + \varphi \nabla W_c \cdot (\mathbf{x} - \mathbf{x}_c),$$

where $\bar{W}_c$ is the cell average component of the state vector (ρ, u, v, p) in V_c, ∇W_c is the estimated gradient on cell V_c and φ is a limiting function, $\mathbf{x}_c = (x_c, y_c)$ is the centroid of the cell V_c.

Several methods have been proposed to approximate the linear gradient ∇W_c. Barth and Jespersen [4] applied the Green–Gauss theorem to compute the gradient within a cell using the cell-center values of its neighbors [34,32]. Here we use the least-square reconstruction to give the gradient approximation. The advantage of such a procedure is more robust and less sensitive to grid irregularity [34,32],

$$\nabla W_c = \mathcal{A}^{-1} \begin{bmatrix} \sum_{f \in \mathcal{F}(c)} (x_f - x_c)(\bar{W}_f - \bar{W}_c) \\ \sum_{f \in \mathcal{F}(c)} (y_f - y_c)(\bar{W}_f - \bar{W}_c) \end{bmatrix}, \tag{57}$$

where $\mathcal{A}$ is 2×2 matrix

$$\mathcal{A} = \begin{bmatrix} \sum_{f \in \mathcal{F}(c)} (x_f - x_c)^2 & \sum_{f \in \mathcal{F}(c)} (x_f - x_c)(y_f - y_c) \\ \sum_{f \in \mathcal{F}(c)} (x_f - x_c)(y_f - y_c) & \sum_{f \in \mathcal{F}(c)} (y_f - y_c)^2 \end{bmatrix}.$$

We shall introduce a classical limitation procedure for the slope in order to achieve a monotonic piecewise linear reconstruction. There are different limiting functions, such as Barth limiter [4] and Venkatakrishnan limiter [45]. A general formulation of a slope limiter can be expressed as follows [37],

$$\varphi = \min_{\forall q \in \mathcal{Q}(c)} \begin{cases} \phi(r_{q,c}^{max}), & \text{if } \nabla W_c \cdot (\mathbf{x}_q - \mathbf{x}_c) > 0, \\ \phi(r_{q,c}^{min}), & \text{if } \nabla W_c \cdot (\mathbf{x}_q - \mathbf{x}_c) < 0, \\ 1, & \text{if otherwise}, \end{cases} \tag{58}$$

where $r_{q,c}^{min}$ and $r_{q,c}^{max}$ are the ratios of the allowable variation to the estimated variation at the vertex q,

$$r_{q,c}^{min} = \frac{\min_{f \in C(q)} \bar{W}_f - \bar{W}_c}{\nabla W_c \cdot (\mathbf{x}_q - \mathbf{x}_c)}, \qquad r_{q,c}^{max} = \frac{\max_{f \in C(q)} \bar{W}_f - \bar{W}_c}{\nabla W_c \cdot (\mathbf{x}_q - \mathbf{x}_c)},$$
$$\phi(r_{q,c}^{min}) = \min(1, r_{q,c}^{min}), \qquad \phi(r_{q,c}^{max}) = \min(1, r_{q,c}^{max}).$$

6.1. *Description of the algorithm procedure*

Now, we are in a position to give a summary of our high order Eulerian method.

Step 1. Initialization

At time $t = t^n$ we know all flow variables in each cell: ρ_c^n, $\mathbf{u}_c^n$, p_c^n, e_c^n, c_c^n and the geometrical quantities: $\mathbf{x}_q^n$, V_c^n, $\mathbf{N}_f^c$, L_f^c. Refer to Fig. 1.

Step 2. Reconstruction

From the cell average state to construct a piecewise linear representation of the state vectors at time t^n over the cell V_c,

$$W_c(\mathbf{x}) = \bar{W}_c + \varphi \nabla W_c \cdot (\mathbf{x} - \mathbf{x}_c).$$

Here $\bar{W}_c = (\rho_c^n, u_c^n, v_c^n, p_c^n)$.

Step 3. Time step limitation

The time step limitation Δt is obtained from (51) for Eulerian approach.

Step 4. Nodal solver

Given the piecewise linear state components $W_c(\mathbf{x}_q), W_{c^+}(\mathbf{x}_q)$ at node q, we employ these quantities as the left and right states across an edge. See Fig. 5. The contact velocity $\mathbf{u}_q^*$ at each vertex is obtained by solving the linear system of equations in (50).

Step 5. The Riemann fluxes of the HLLC-2D solver

Refer to Fig. 2(b). At the common edge $c \wedge f$ between the cell V_c and V_f, define conservation quantities $\mathbf{U}_{L,k}, \mathbf{U}_{R,k}$ from state vectors $W_c(\mathbf{x}_q)$ and $W_f(\mathbf{x}_q)$. The half edge fluxes $\mathbf{F}^c_{f,q}$ and $\mathbf{F}^f_{c,q}$ are calculated from (32) and (33). Repeat this procedure at node q^+. The resulting edge fluxes $\mathbf{F}^c_f$ and $\mathbf{F}^f_c$ are the average of the two half edge fluxes (39) and (40).

Step 6. Update

The conservation quantities in cell V_c are updated by (3). It goes to Step 1 if the calculation continues, otherwise it stops.

7. Numerical experiments

Extensive numerical experiments have been carried out to demonstrate the performance of the newly proposed schemes. These testes use triangular or quadrilateral meshes and all triangular meshes are generated by a Delaunay triangulation algorithm. For the sake of consistency with the previous solution procedure, we use the time discretization with first order accuracy. A second-order accurate time procedures can be obtained by using Runge–Kutta method. In what follows, all test examples adopt CFL number $C_E = 0.3$.

7.1. Shock tube problem

The first test is devoted to a one-dimensional Sod shock tube problem computed on a 2D mesh. Due to the equivalence between the HLLC and the HLLC-2D in rectangular mesh, we only consider the numerical results on an unstructured triangular grid. The computational domain is $[-0.5, 0.5] \times [0, 0.1]$ with a triangulation of 201 vertices in the x-direction and 21 vertices in the y-direction. Initial values are

$$\begin{cases} (\rho_L, u_L, v_L, p_L) = (1, 0.2, 0, 1), & \text{if } -0.5 \leqslant x < 0, \\ (\rho_R, u_R, v_R, p_R) = (0.125, 0.2, 0, 0.1), & \text{if } 0 \leqslant x < 0.5. \end{cases}$$

Fig. 10 shows the one-dimensional density and pressure on a section plane parallel x-axis for the HLLC and HLLC-2D schemes. Both first and second-order accurate numerical approximations are compared to the exact solution. Fig. 11 exhibits the second-order accurate density distributions on the 2D plane.

We can see that two schemes perform well with good resolution. Some oscillations are observed across the contact discontinuity due to start up errors. Comparatively, the new HLLC-2D scheme is slightly more dissipative than the classical HLLC solver but it reduces obviously numerical oscillations appearing in the HLLC calculation.

7.2. Odd-even decoupling

Fig. 12 shows Quirk's odd–even decoupling test problem [38] using the first order scheme and quadrilateral mesh. This test is used to exhibit the stability of the numerical shock. The gas has the specific heat ratio $\gamma = 1.4$. A moving shock of $M_s = 6.0$ propagates down a tube. To the right of the shock, the initial conditions are $(\rho, u, v, p) = (1.4, 0, 0, 1)$. The computational field is 800×20 and the spatial grid step lengths are $\Delta x = \Delta y = 1$. The centerline of the grid is perturbed from a perfectly uniform grid by $\pm 10^{-3}$. We note the magnitude of perturbation is the same as that in [36] and larger than that used in [38] to emphasize the problem of shock instability. The initial position of the shock wave is at $x = 20$. The results show the density contour at $x \approx 300$. The HLLC scheme shows the promotion of odd–even decoupling along the shock, whereas the results coming from the HLLC-2D solver show that decoupling is completely eliminated.

7.3. Mach 20 hypersonic flow over a cylinder

This is a well-known test to examine the catastrophic carbuncle failings of upwind schemes. We need to simulate an inviscid flow with Mach number $M_a = 20$ around a circular cylinder. In this test problem, the specific heat ratio of the gas is $\gamma = 1.4$, and the initial inflow has states $(\rho, u, v, p) = (1, M_a\sqrt{\gamma}, 0, 1)$. The computing domain is displayed in Fig. 13 and 40×400 structure grids and the first order accurate scheme are used. The density contours at $t = 1$ by the HLLC and HLLC-2D solvers are illustrated in Fig. 13. The carbuncle phenomenon appears when using the HLLC scheme and disappears when using the HLLC-2D scheme.

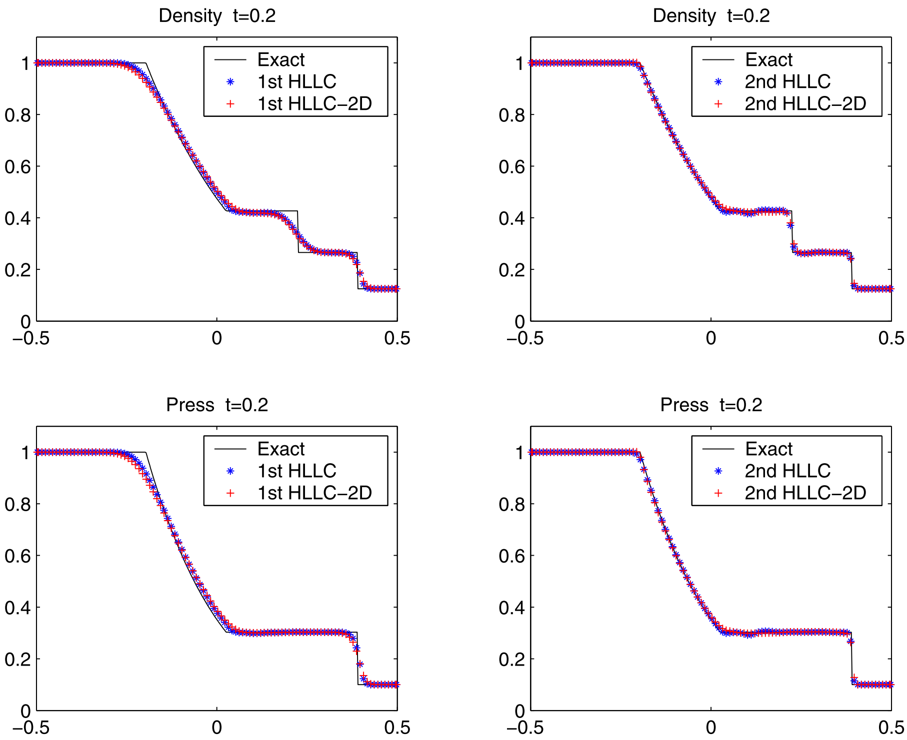

Fig. 10. Comparison between the HLLC and HLLC-2D solvers in the Sod problem.

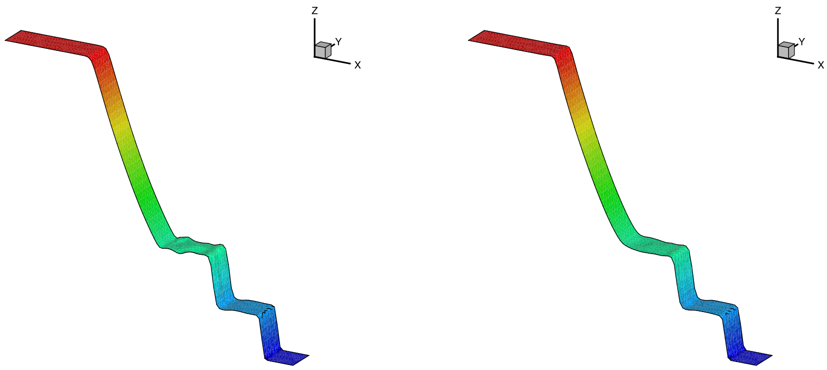

Fig. 11. Density plots in the Sod problem. Left: HLLC, right: HLLC-2D.

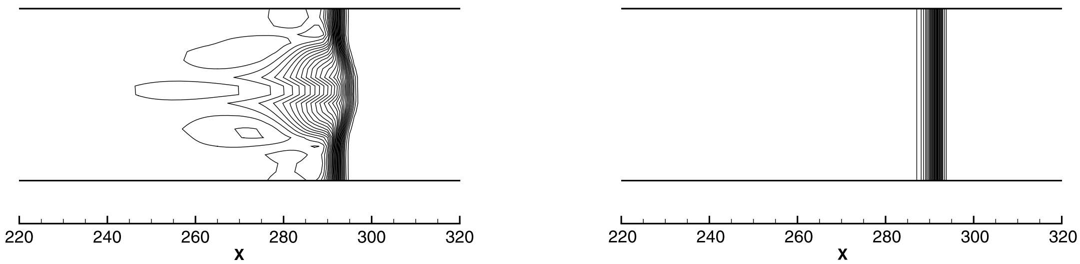

Fig. 12. The density contours for the odd–even grid perturbation problem. Thirty equally spaced contour lines from $\rho = 1.6$ to $\rho = 7.6$, left: the HLLC, right: the HLLC-2D.

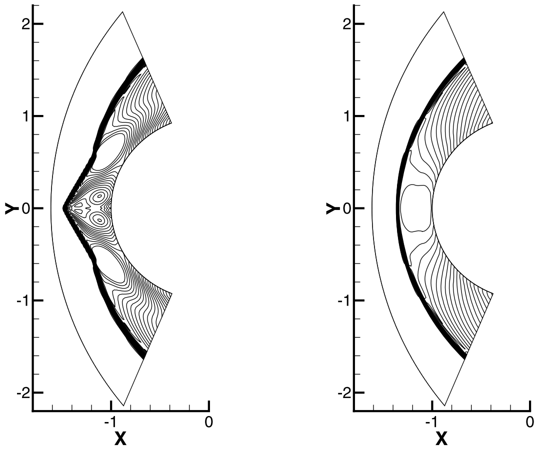

Fig. 13. The density contours for a hypersonic flow over a cylinder. Thirty equally spaced contour lines from $\rho = 1.2$ to $\rho = 6$, left: HLLC, right: HLLC-2D.

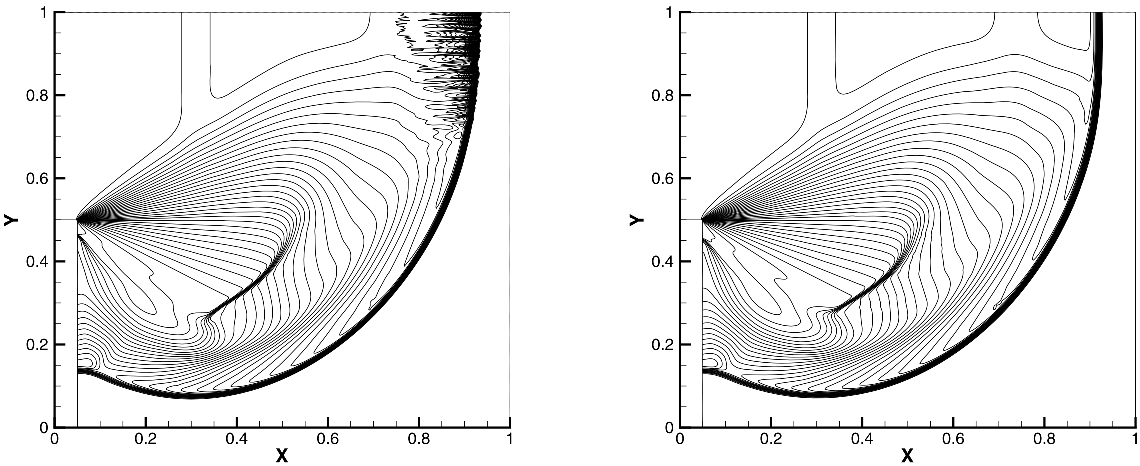

Fig. 14. The diffraction of a supersonic shock moving over a 90° corner. Thirty-five equally spaced density contour lines from $\rho = 0.2$ to $\rho = 7$, left: HLLC, right: HLLC-2D.

7.4. The diffraction of a supersonic shock moving over a 90° corner

This example shows a moving shock with Mach number 5.09 diffracts around a 90° corner. The computational domain is a unit square $[0, 1] \times [0, 1]$ that is discretized into a $20 \times 200 + 400 \times 400$ quadrangle cells. The corner is at $(x, y) = (0.05, 0.5)$. Initially, the shock is at $x = 0.05$. To the right of the shock, the flow field is initialized to

$$(\rho, u, v, p, \gamma) = (1.4, 0, 0, 1, 1.4).$$

Quirk [38] has pointed out that some Riemann solvers might encounter difficulties on a fine mesh. Fig. 14 gives the density contours obtained by using the first order HLLC and HLLC-2D. The HLLC scheme shows spurious oscillations at the planar moving shock and HLLC-2D solver has good shock picture.

7.5. Kinked Mach stem

This problem was studied extensively by Woodward and Colella [47]. Initially, a Mach 10 shock is located at $x = -0.1$ and travels in the x-direction and encounters a wedge that makes an angle 30° with the shock propagation. The undisturbed air ahead of the shock has a state $(\rho, u, v, p) = (1.4, 0, 0, 1)$. The computational domain is $[0, 2.4] \times [0, 2.1]$ and computations run up to the dimensionless time $t = 0.2$. In this case, a double-Mach reflection (DMR) is formed. For some schemes the primary Mach stem is sometimes kinked, similar to the carbuncle phenomenon. We use second order scheme on unstructured mesh in this example. Fig. 15 gives the density contours computed on a triangular mesh with 218 202 cells. The HLLC scheme shows the kinked primary Mach stem and instable result behind the shock. Using the new algorithm, the spurious kinked Mach stem and instability near the boundary are eliminated.

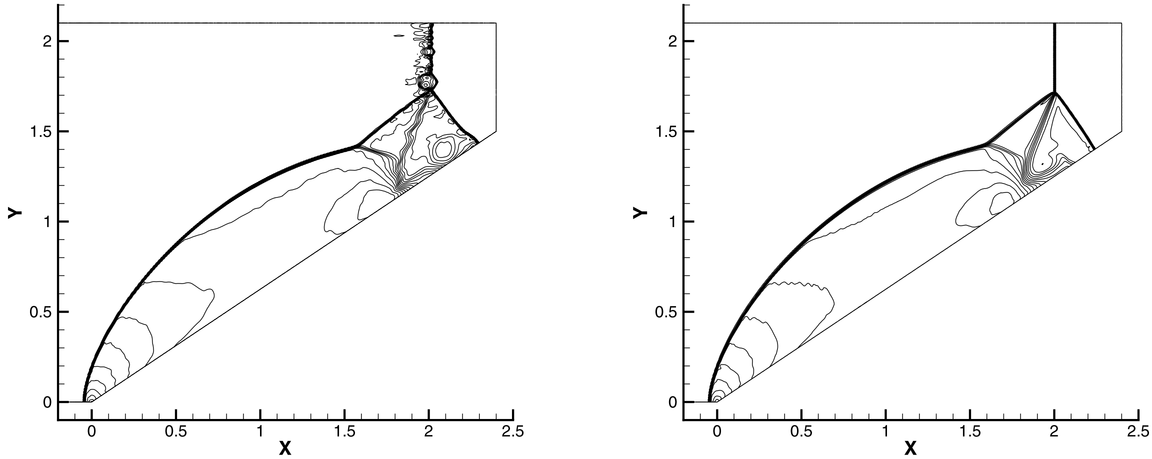

Fig. 15. Kinked Mach stem problem. Thirty equally spaced density contour lines from $\rho = 2$ to $\rho = 22$, left: HLLC, right: HLLC-2D.

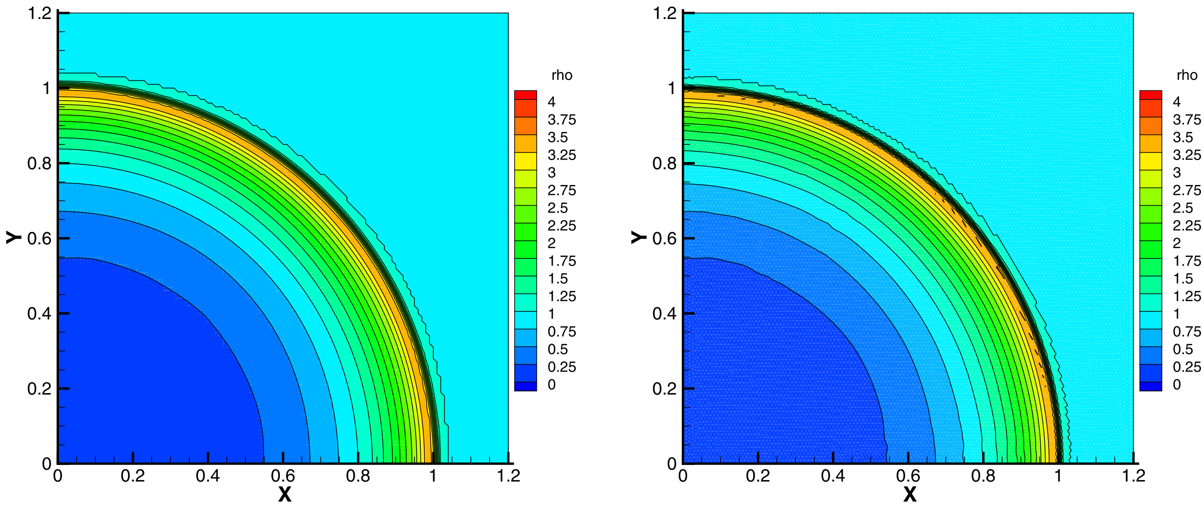

Fig. 16. The density contours of the 2nd order accurate Sedov results at $t = 1$. Left: the quadrilateral mesh, right: the triangular mesh.

7.6. Sedov problem

Sedov [41] obtained an analytic solution for a blast wave that is generated by a point source. We model this problem on a domain $[0, 1.2] \times [0, 1.2]$ with both quadrilateral and triangular meshes. The quadrilateral mesh adopts 120 cells in each direction, and the Delaunay triangular mesh keeps the same mesh size with the rectangular one, where the mesh size is taken to be the maximum diameter of the circumcircles of the triangles. The cell number is 34 188 and the vertex number is 17 337. The initial state is

$$(\rho, p, u, v, \gamma) = \left(1, 10^{-6}, 0, 0, 5/3\right),$$

A finite source internal energy at the origin is $e_0 = 0.564113$. It will produce a shock that is located at radius 1 at time $t = 1$. The specific internal energy near the origin is initialized with 1 cell for the quadrilateral mesh and two cells for the triangular mesh. The values are $e_0/(4|V_c|)$, where $|V_c|$ is the sum of the cell volumes connected to the origin, and the factor 4 represents our calculation is on the first quadrant.

In this test example, we use both the 1st order and 2nd order approaches on the quadrilateral and the triangular meshes. Fig. 16 compares the density contours on quadrilateral and the triangular meshes. The density and pressure fields on the two meshes are plotted in Figs. 17 and 18 respectively. Notice that physical quantities in each cell appear in the latter two figures to exhibit the symmetry of the numerical results. Clearly, both the first order and second order accurate results on the two meshes are well symmetric, and the 2nd order approach has better accuracy and give a good agreement with the analytic solutions. Although the triangular mesh is not symmetric, the results still keep the symmetry rather well.

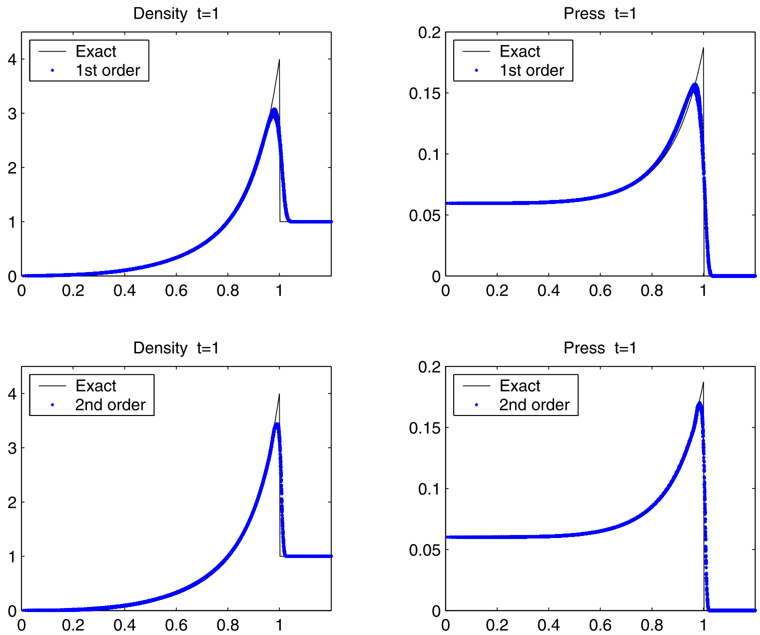

Fig. 17. The Sedov results using the quadrilateral mesh at $t = 1$. Top: the 1st order solution, bottom: the 2nd order solution. Every cell center value in the mesh is plotted.

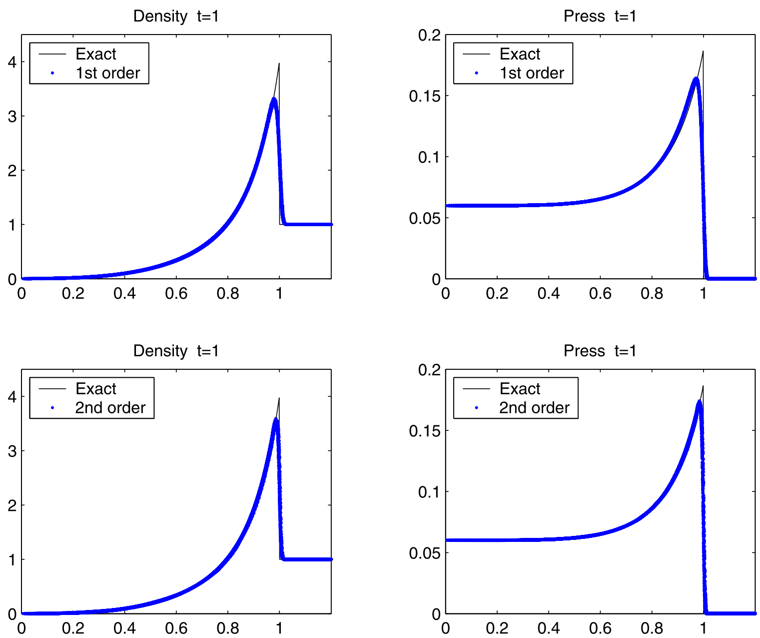

Fig. 18. The Sedov results using the triangular mesh at $t = 1$. Top: the 1st order solution, bottom: the 2nd order solution. Every cell center value in the mesh is plotted.

7.7. Triple point problem

Here, we consider a mono-material triple point problem which corresponds to a three-state two-dimensional Riemann problem. The computational domain $\Omega = [0, 7] \times [0, 3]$ is split into three regions, and the regions and states at the initial time are

$$\Omega_1 = [0, 1] \times [0, 3]: \quad \rho_1 = 1,\ u_1 = 0,\ v_1 = 0,\ p_1 = 0.8,\ \gamma = 1.4,$$
$$\Omega_2 = [1, 7] \times [0, 1.5]: \quad \rho_2 = 1,\ u_2 = 0,\ v_2 = 0,\ p_2 = 0.1,$$
$$\Omega_3 = [1, 7] \times [1.5, 3]: \quad \rho_3 = 0.125,\ u_3 = 0,\ v_3 = 0,\ p_3 = 0.08.$$

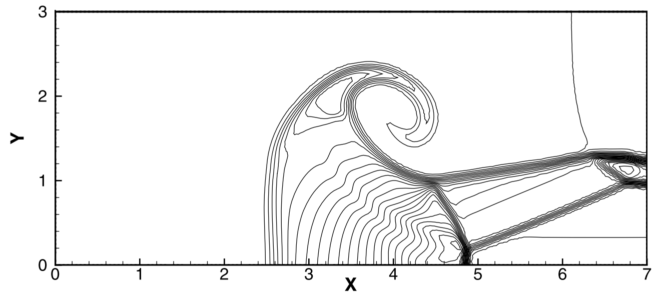

Fig. 19. The density field of the second order Lagrangian HLLC-2D approach on a triangular mesh in triple point problem, $t = 5$. Forty equally spaced density contour lines from $\rho = 0$ to $\rho = 5$.

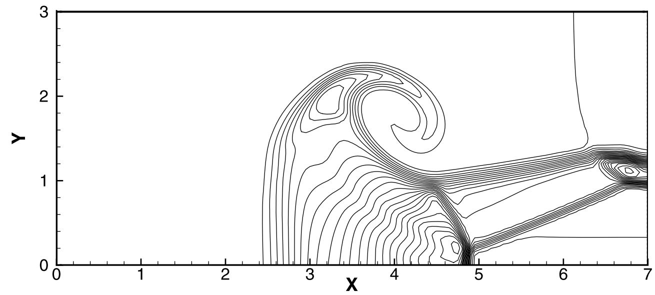

Fig. 20. The density field of the second order Lagrangian HLLC-2D approach on a quadrilateral mesh in triple point problem, $t = 5$. Forty equally spaced density contour lines from $\rho = 0$ to $\rho = 5$.

The triple point problem is used to assess the robustness of a Lagrangian method on a problem that has significant vorticity [23]. The left region Ω_1 has a high pressure that drives a shock through two connected regions $\Omega_2 \cup \Omega_3$ and a vortex develops at the triple point where three regions connect. We use quadrilateral and triangular meshes and the Lagrangian HLLC-2D method to simulate this problem. The grid resolution is 140×60 on a quadrilateral mesh. The triangular mesh has the same mesh size with the rectangular one. The cell number is 19 806 and the vertex number is 10 104.

Figs. 19 and 20 show the second order solution using triangular and quadrilateral meshes at $t = 5$. In Eulerian calculation, the results on the triangular mesh have better accuracy than those on a quadrilateral mesh. It is because the triangular mesh has more grid cells. The stiffness phenomenon caused by triangles in Lagrangian calculation does not appear [42].

8. Summary

We have developed a new cell-centered conservative scheme. This scheme includes a genuinely two-dimensional Riemann solver, HLLC-2D, based on cell states around a node. The main new features of the algorithm are the introduction of the nodal particle motion velocity and discontinuous fluxes on each side of the edge. The nodal contact velocity is determined by the conservation of mass, momentum and total energy. In the procedure of deriving numerical flux, we relax one-dimensional contact pressures in traditional HLLC solver to ensure the consistency between the flow velocity on an edge and the nodal contact velocity. In order to obtain the advection terms we introduce two approximate fluxes on each initial state to ensure Rankine–Hugoniot conditions are satisfied for each flux. The numerical flux on the edge of cell is compose of the two fluxes. The resulting scheme can be interpreted as the two-dimensional generalization of HLLC solver or a Eulerian extension from Lagrangian formalism proposed by Maire et al. In order to improve the accuracy of the scheme we have developed its second order extension with the MUSCL type method.

In the alternative HLLC-2D formulation, there are obvious contact wave, therefore it is contact preserving. Moreover it appears to be quite robust according to the numerical results obtained for the various test cases presented in this paper. Due to the multi-dimensional properties of the HLLC-2D solver, the numerical shock instability is eliminated effectively.

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